

SVI • CAS

Rev 6.0

Digital Tape Drive Playback & Recording System Owners Manual

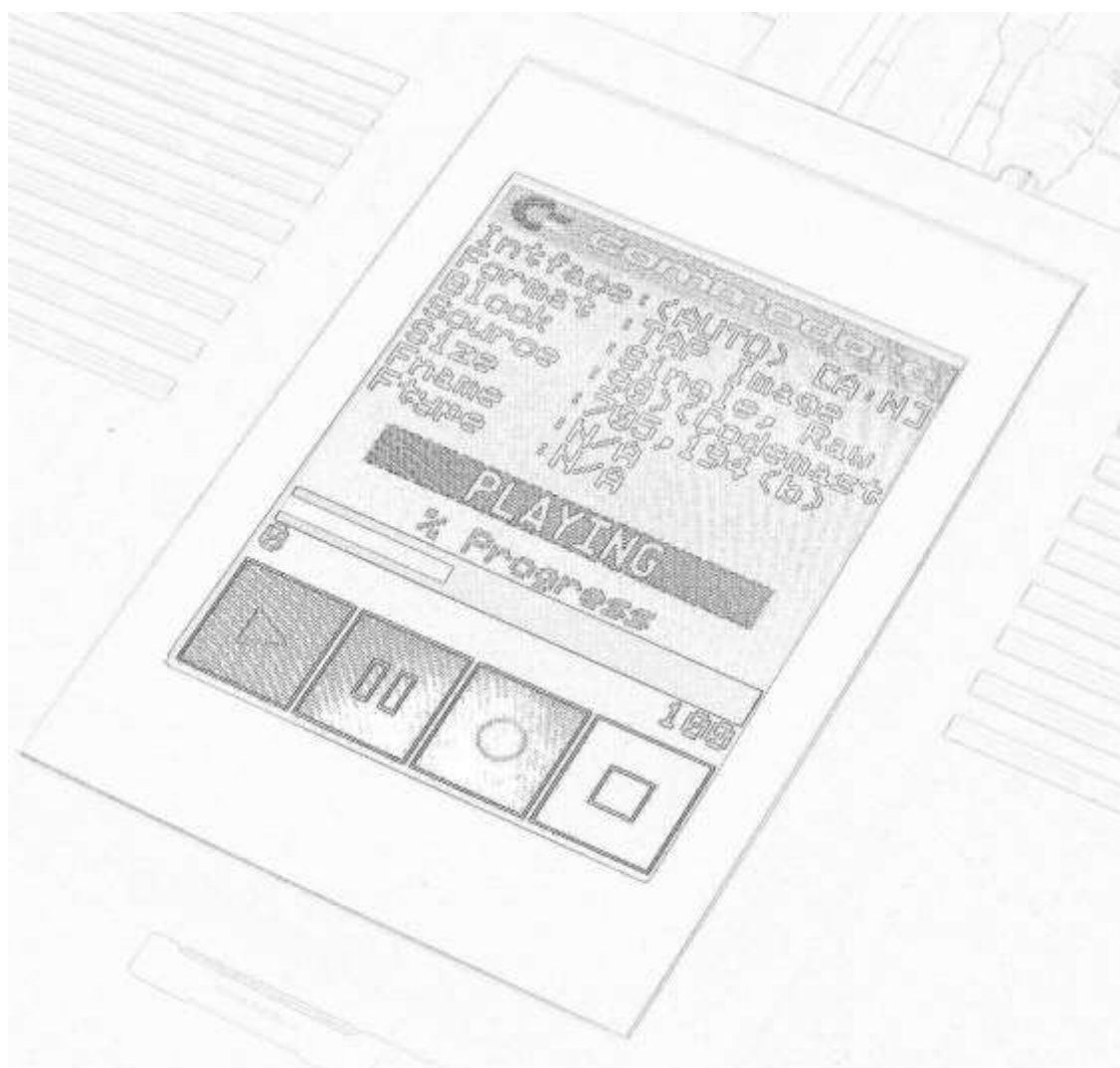


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1. INTRODUCTION

SVI-CAS is the successor to the now discontinued **SVI-WAVE**.



Figure 1.1. SVI-WAVE

Whilst **SVI-WAVE** provisioned a means to load **CAS** converted WAV files, it still required the services of third party equipment to actually play the source image into the host SVI-318/328 computer. Accordingly, the manual loading of wave files into these vintage machines still remained a some-what cumbersome and unreliable operation at the best of times.

SVI-CAS addresses this short-fall by providing SVI-318 / 328 owner with a fully self-contained SD card based cassette image playback and recording system for the very first time. In addition, the author has included features and operational capabilities to support additional computers that can't be found in any of the existing known image players at time of writing.



Figure 1.2. SVI-CAS

SVI-CAS key features:

- **Full SD card based image playback & recording system**
- **SD card class 10 card compatibility**
- **Colour TFT touch screen driven**
- **No playback or recording level adjustments – plug & play design**
- **Fully configurable audio feedback**
- **Baud rate and tape image play-back control – machine dependent**
- **Proper directory navigation & file deletion facilities**
- **Image inspection / internal file selection launching**
- **Playback position memorization for later repeat playback**
- **Adapter cables allow Atari, SVI-318/328 & CBM cassette use**
- **Range of image types supported, e.g. CAS, TAP, TSX, TZX, and so on**
- **Support for Atari, Spectravideo, TRS-80 CoCo, Dragon, MSX, Sinclair, Amstrad, Acorn, Commodore, Oric, & most other retro systems**

1.1. Technicalities

SVI-CAS commenced development within a TI MSP430 environment, but ended up being based upon the hardware of the MEGA 2560 project development board paired with a colour TFT touch screen for cost reasons. The firmware was written from scratch in a combination of the C programming language and Atmel assembler under the Codevision AVR IDE & compiler, debugging being via AVR Studio 7 and an Atmel ICE.

This is not an **Arduino** based product and makes no use of any source code or design aspects of the known existing public domain image player projects. This product does not make use of a boot-loader. The **SVI-CAS** firmware and hardware design, plus fundamental operational characteristics are alien to all existing Arduino designs.

1.2. Contents

SVI-CAS when purchased, may come with one or more special adapter cables for connection to either Atari, Spectravideo SVI-318 / 328, or Commodore 8-bit colour computers as well as this manual. It is suggested the user reads this manual completely before attempting to use the product and hence reduce the risk for potential problems.



Figure 1.2.1. SVI-CAS with CBM and SVI Adapter cables

Should the user have questions or comments / suggestions they are welcome to forward then to the author via the purchase channel used to obtain **SVI-CAS**.

1.3. Requirements

SVI-CAS requires the user to supply a number of items prior to use, these being:

- An SD card, 2GB is considered to be of a suitable size, larger cards are fine as well, though an entire library size of images for all supported vintage computers are only a fraction of the storage space such cards offer. The only exception to this general rule is in respect of raw TAP images for Commodore machines, as libraries of such images may be as large as several gigabytes on their own.
- A mains plug-pack style 9V DC @ 400mA (or greater), (center +, outer -) regulated adapter is required to power **SVI-CAS**. The plug pack connector should be a 2.5mm power jack that is at least 5mm in length.

Ensure that plug-pack is a modern switch-mode unit – older transformer based adapters tend to exhibit output voltages that can reach more than 9V and will cause SVI-CAS to cut-out due to over-heating of the internal voltage regulators!

If in doubt, use a multi-meter to measure the output voltage and verify correct 9V regulation is present. **SVI-CAS** is also diode protected, so application of a reverse polarity voltage (opposite polarity plug-pack) will not damage it.

A good indication as to a mains adapter being of the switch-mode variety or old-school transformer type is the weight, a switch-mode variety adapter will feel like the case is actually empty – very light, while a transformer based unit will feel heavy when held in your hand. Avoid the transformer type due to their higher than stated voltage level output.

- Lastly for both TRS-80 CoCo, MSX, Amstrad, Acorn, & Sinclair use it is strongly urged that the user either buy or make a suitable cables for connecting **SVI-CAS** to their computer, as the user will tire very quickly of manual operation when loading images that contain many files. Computer control not only preserves user sanity, but it really allows the computer to control playback properly and reduces the risk of loading failures.

PLEASE NOTE: Earlier versions of SVI-CAS required use of a 12V DC mains adapter, as of revision 4.0 of this product ONLY use a 9V DC mains adapter, this change has occurred through the removal of the ability to power this device directly by an SVI-318 or SVI-328 computer, which was rarely used by most owners.

1.4. Supported Image Formats

SVI-CAS supports several cassette image formats for a number of machines:

- **Atari 400 / 800 / XL / XE Series**
 - CAS
- **SVI-318 / 328**
 - CAS
 - TSX
- **TRS-80 CoCo / MC10**
 - CAS
- **Dragon 32 / 64**
 - CAS
- **MSX machines**
 - CAS
 - TSX
- **Sinclair ZX80**
 - OZX ('.O' suffix images)
 - TZX
- **Sinclair ZX81**
 - PZX ('.P' suffix images)
 - TZX
- **Sinclair ZX Spectrum**
 - TAP
 - TZX
- **Amstrad CPC machines**
 - CDT
- **Acorn machines**
 - UEF
- **Commodore machines**
 - TAP
- **Oric-1 & Atmos machines**
 - TAP
- **Generic mode**
 - WAV (RIFF) recordings
 - TZX generic images
 - SVI-CAS TAP recordings

Interface selection instructs **SVI-CAS** how to behave and treat image files in respect of the machine's abilities that **SVI-CAS** is connected to, hence proper interface selection is crucial for sensible operation. Improper selection may lead to dysfunctional unproductive behavior.

1.5. SD Card Preparation

Prior to use prepare a blank FAT32 formatted SD card and load any collections of desired image files as required for the machine(s) the user intends support with **SVI-CAS**.

Gently install the SD card face-up within **SVI-CAS**, in the event of difficulty or a jammed card, the SD card socket may be fully exposed via careful removal of the bottom access panel to facilitate removal. Slide the access panel forward slowly and stop when it protrudes about 1cm, then lift it away at an angle being careful NOT to damage the small ribbon cable. Refit panel in the reverse manner.

SD cards are hot-swappable with **SVI-CAS**, just be sure to return to the main menu first to avoid confusing **SVI-CAS** – then gently remove the current card and insert another. All used cards must be pre-prepared to be FAT32 compliant.



Figure 1.5.1. SD card slot

The SD card slot is located at the base of the image player as shown in the above image, it is a full-size SD slot, thus is much easier to handle and swap cards than with smaller card interfaces. At any point during SD card reading should **SVI-CAS** experience a read-error, it will announce this via the following message:



SD Read Error

Figure 1.5.2. SVI-CAS Indicates SD Reading Problem

1.6. Touch Screen Operation

SVI-CAS has no physical buttons and is driven via its touch screen, only reasonable pressure is required to register a button press, avoid the use of sharp objects or significant pressure when using the touch screen to avoid the risk of damage. The use of fingers (with short nails) is considered the best option with the buttons having been deliberately sized to allow soft finger presses to work well in driving the unit. A plastic touch screen stylus is included with this unit which may also be used, the above mentioned precautions still applied. The user should always ensure their hands are clean and free of contaminants such as grease and oil when using this unit, such contaminants may cosmetically permanently stain the image player.

Only wipe the touch screen with a fine soft dry cloth gently if needed, never apply cleaning solutions to this unit as both cosmetic and functional damage may occur.

During use the TFT display will become warm, there should be no cause for alarm, this warmth is due to the back-light LEDs of the TFT display. Some additional heat will be derived from internal voltage regulation.



Figure 1.6.1. 2.8" TFT & Touch Screen Display & Piezo Speaker (Rear of case)

The rear of the unit has a hole to allow audio generated by the internal piezo speaker to be heard, ensure that this outlet is not obstructed and never insert objects as this can damage the speaker.

2. THE DISPLAY

During the various stages of operation **SVI-CAS** will presents the user with information visually, which is intended to assist in the preparation and actual play-back of program images. This section documents the various menus & screens that will be encountered as well as their content variations and associated meanings. The order in which screens are presented is the general order in which **SVI-CAS** operates under normal conditions.

2.1. Start Up

At initial power-on **SVI-CAS** will display a splash screen, describing both the product name and a software version number (not visible in example figure) as shown in the example of **Figure 2.1.1.** below:



Figure 2.1.1. Splash Screen

A short multi-note tune will sound and after several seconds the display will progress to the main menu.

During this time **SVI-CAS** will also initially check for the presence of a working SD card as part of self testing, should a working card not be identified an error message will be generated together with the displaying of the following warning symbol in **Figure 2.1.2.** :



Figure 2.1.2. SD card Check-Fail Symbol

At this point **SVI-CAS** will halt and require the user to power down the unit and either insert a correctly formatted and image-loaded SD card, or inspect any present SD card for possible problems such as incorrect or incomplete insertion. Successful SD card confirmation will result in **SVI-CAS** progressing to the default main menu.

SD cards should also be checked for broken plastic guide fingers which can obstruct proper electrical connection and lead to failed loading operations, never attempt to force a card with broken or bent fingers into the card reader lot, as this may permanently damage the internal card reader.

2.2. Main Menu

After a successful power-up **SVI-CAS** always defaults to its main menu as shown in **Figure 2.2.1.** below:



Figure 2.2.1. Main Menu

The main menu consists of three distinct areas, the first specifies the current interface selected based upon the host computer, the second is a collection of attributes related to that specific interface, while the last provides buttons for accessing the operational functions of **SVI-CAS**.

Note that the displayed preferences may differ according to what was previously selected before **SVI-CAS** was previously powered down, the settings hence are not key here, but that the main menu is visible and has the above general visual representation. The main menu serves as the means of configuration prior to program image play-back.

Options may be described via text or depicted as graphical images, the latter appearing in one of two possible colours, white meaning that this option is active, or red denoting that the options is currently disabled. Taping on these respective buttons will cycle between possible settings. All possible options that are depicted graphically are shown even when disabled, it is only text based settings that cycle on-by-one when those respective buttons are pressed,

It is important for the user to become familiar with these options as they may or may not apply to a specifically selected interface (computer). Improperly set options can lead to **SVI-CAS** not correctly playing back a selected image

There are seven buttons comprising the main menu, their functionality is described in the table(s) that follow.

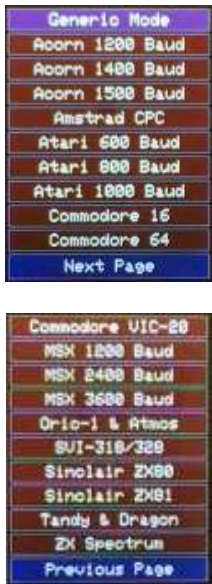
Function	Options
Interface Selection (via multi-page list) 	• Generic Mode • Acorn 1200 Baud • Acorn 1400 Baud • Acorn 1500 Baud • Amstrad CPC • Atari 600 Baud • Atari 800 Baud • Atari 1000 Baud • Commodore 16 • Commodore 64 • Commodore VIC-20 • MSX 1200 Baud • MSX 2400 Baud • MSX 3600 Baud • Oric-1 & Atmos • Spectravideo SVI-318 / 328 • Sinclair ZX80 • Sinclair ZX81 • Tandy & Dragon • ZX Spectrum

Table 2.2.1. Interface Selections

When selecting a different computer interface SVI-CAS will automatically adjust certain parameter settings to match the selected host machine, this removes the need for the user to remember the optimal settings for every host machine they may use. Any machine used in generic mode will be subject to uniform default settings and will require final adjustments by the user for optimal performance.

Baud Rate Control	•  Automatic Baud (managed by SVI-CAS) •  Baud Manual (user setting is applied for those machines that allow adjustment)
Recording Edge Trigger	•  Trigger on rising edge •  Trigger on falling edge • If both selected then triggering on both edges applies (available for non-CBM machines only)
Play-Back Control	• White is enabled, red is disabled • If both disabled then image plays from start to finish without user or computer intervention. •  computer control •  manual control
Timing Control	• White is enabled, red is disabled • Red denotes function disabled • PAL for European Commodore machines • NTSC for North American Commodore machines

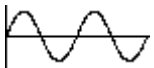
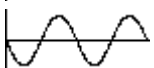
Audio Options	• White is enabled, red is disabled • Red denotes disabled •  menu operational sounds •  image playback audio
Waveform Phase	• White is enabled, red is disabled •  Output signal is non-inverted (0°) •  Output signal is inverted (180°)
FILES	• Opens the file directory view of the SD card
RECORD	• Opens the file name entry menu for recording

Table 2.2.1. Main Menu Attributes & Functionality.

2.3. Directory Menu

The directory menu provides a means for the user to navigate through both directories and file collections in order to select a desired image for inspection and play-back or to delete it if desired



Figure 2.3.1. Sample Directory Display

The directory display is split, with the upper portion of the display dedicated to presenting the names of both directories and files based on the user's current navigated position within the SD card file allocation table.

SVI-CAS will allow the user to drill down to a maximum of eight nested-levels of directory, while there is no enforced limitation on the number of files that may reside in a given directory – though it must be stated that an excessive number of files will degrade speed of navigation. It is also important for the user to understand that the order in which directory names and files are written to an SD card will affect overall access performance.

A total of eleven file or directory names (mixed or un-mixed) are displayable within the directory screen at any one time, the current selected file or directory name being highlighted with a red background. A maximum of up to twenty characters for each name are displayed within each row, should a name be longer than this, after a period of five seconds the name will begin to scroll a character at a time from right to left to allow the full name to be accessible.

Key Points:

- **SVI-CAS** supports 255 character directory and file names
- A directory name is displayed as green text using the format “< name >”
- A file name is displayed as white text with no special format
- File names longer than 20 character will scroll after five seconds
- Directory names longer than 18 characters will scroll after five seconds
- The user may elect to delete files, but not directories

A typical directory display is depicted as follows in **Figure 2.3.2.**



Figure 2.3.2. Directory Menu

The user may have up to 65,535 files within a directory which are reachable by **SVI-CAS**, though it is strongly urged to limit file counts to less than 200 to avoid severe slowing of **SVI-CAS** operation. File chains are read sequentially from serial devices such as SD cards and USB sticks, so it is suggested to apply common sense in regards to directory creation and usage and avoid excessive numbers of files within individual directories.

Table 2.3.1. which follows, describes the button-based navigation of this particular menu level.

Function	Options
	Will return the user to the main menu
	- If a directory is highlighted this will drill down to the next directory level (maximum of eight nested levels) - If a file name is highlighted this will move to inspect the image contents
	Delete the selected file (displays in white), deletion of directories is not permitted (displays in red)
	- If within a sub-directory, this will drill up one directory level - If already at the root level of the directory, this will drop the user back to the main menu
	Move back one page of entries (if possible)
	Move back one displayed page entry (if possible)
	Move down one displayed page entry (if possible)
	Move forward one page of entries (if possible)

Table 2.3.1. Directory Menu Navigation Functionality

When navigating forwards or backwards through complete pages of entries, **SVI-CAS** will attempt to maintain the current selected row, only adjusting where the next page of entries does not have an entry for that row, instead selecting the last available entry in that case.

It is also important to remember that **SVI-CAS** does not pre-sort directory and file names prior to display, hence the order presented will represent the order in which said directories and files were originally created upon the installed SD card. Should a selected directory be empty, **SVI-CAS** will advise so (see below) upon the user attempting to open this directory, also disabling the  button from further use (shown as red).



Directory Empty

Figure 2.3.3. Directory Empty

2.3.1. Deleting Files

The directory menu allows for the deleting of files should the user no longer require them, to delete a file select it using the navigation buttons and then press the delete button, **SVI-CAS** will then prompt the user to confirm they would like to delete the file. Selecting the confirm option will result in the file being removed while the cancel option will abort the operation.

SVI-CAS does not permit the deletion of directories, should such maintenance be required the SD card should be transferred to a PC in order to remove directories and any subsequent files or sub-directories. After deleting a file **SVI-CAS** will show the updated directory post file removal.

2.4. Image Inspection Menu

The image inspection menu allows for the user to both examine the contents of a program image whilst also selecting a contained entry for the commencement of play-back into a supported computer.



Figure 2.4.1. Sample Image Inspection Display

SVI-CAS supports program images that contain a maximum of 40 internal entries in terms of being able to browse and select for play-back, any additional entries will be ignored, thus larger image collections may still be played-back, but must start from a selection made from the first 40 entries. Each displayed entry is also described in terms of file type to assist the user in loading a selected entry syntax-wise.

Recognized (Header) Entry Types:

- **SVI-318 / 328**
 - ASCII
 - SVI BASIC
 - Binary
- **TRS-80 CoCo / MC10**
 - BASIC
 - Binary
 - Data

- **Dragon 32 / 64**
 - BASIC
 - Binary
 - Data
- **MSX**
 - ASCII
 - BASIC
 - Binary
- **ZX Spectrum**
 - BASIC
 - Binary
 - Character Array
 - Number Array
- **Amstrad CPC machines**
 - Internal BASIC
 - Binary
 - Screen Image
 - ASCII
- **Acorn machines**
 - Headers contain no file-type indicator and are simply generically labeled 'Acorn'
- **Oric-1 / Atmos**
 - BASIC
 - Binary
 - Custom

Wave images, Atari, Commodore, ZX80/81 and **SVI-CAS** TAP recordings use image formats that do not provide **SVI-CAS** with information that allow the internal dissecting of files within an image for these machines.

SVI-CAS will not recognize custom file entries within an image, only entries that are tape BIOS compliant for the respective machine based on the main menu interface selection will be identifiable. This does not mean that **SVI-CAS** cannot play-back custom images, it simply implies that a standard machine BIOS compliant format is required for any selected entry as a playback starting point (other than the very start of an image). There are two play-launching options depending of the composition of the image being played, in most cases the  option will suffice, but should an image start with custom data blocks and then have another program with a recognized header, it may be necessary to use the  option to force playback to begin with those custom blocks first.

In the event that **SVI-CAS** cannot identify any standard image entries it will respond with a warning (see below), but will still allow the attempted playback of the image unless no actual form of starting point at all can be deduced for the expected image structures (based on interface selection).



Figure 2.4.2. No (Bios Compatible) Headers Found

Often later game programs produced for MSX computers as well as the Sinclair ZX Spectrum consisted of two-sided tape recordings, the first side held the actual game program – while the second held all the data for carious levels in a game. It is usually the second side, or in this case the second image of a multi-image game that is of a custom ‘header less’ nature.

It should also be noted that flawed or non-image files will also produce this result when searched for machine-compliant headers. A typical image inspection display is depicted as follows in **Figure 2.4.3.**



Figure 2.4.3. Image Inspection Menu

Should **SVI-CAS** encounter a file name that is either nothing but spaces or simply a null character, it will display this entry with the message “*no name*” to indicate that a file exists but has no visible file name.

The table which follows describes the button-based navigation of this menu level.

Function	Options
	Return the user to the main menu
	Start play-back of currently selected image from very start of overall file ignoring any selected entry
	Start play-back of currently selected image from the selected entry
	Go back one step or directory level

	• Move back one page of entries (if possible)
	• Move back one displayed page entry (if possible)
	• Move down one displayed page entry (if possible)
	• Move forward one page of entries (if possible)

Table 2.4.1. Image Inspection Menu Functionality

In the prior example it can be seen the selected image contains at least 8 entries (maybe more) which are a mixture of ASCII, BASIC, and Binary files. Navigation is the same as the Directory Menu, but is limited to a maximum of 5 pages of up to 8 entries per page. The selected entry is highlighted with a red background and will be used for play-back if  is pressed.

In the event that the selected image file is found to be corrupted or proves difficult to accept based on the expected image type(s), **SVI-CAS** will disable the  button and change the button text colour to **RED**. It is not always possible to identify an image file as flawed or not even an image file, especially for “**CAS**” images therefore the user should ensure they acquire such images from a reputable source to save time and frustration. When searching a source image for entries, **SVI-CAS** will take longer for images that are larger compilations of games for example, so please be patient, a file size of several hundred K-bytes can take several seconds to search completely. **SVI-CAS** will display an hour glass while it is busy searching a file.



Figure 2.4.4. SVI-CAS Busy, Please Wait

The user will note that the red ‘selection-band’ will vary in size according to the selected machine interface – this is due to the fact that standard header file names vary in size according to the machine, SVI & MSX computers use 6 characters, while TRS-80 CoCo uses 8, and the ZX Spectrum uses 10. In addition to this the user will note that some names appear rather strange – there is nothing wrong, these are just the naming conventions that the program authors chose as the time of writing their software, they are just normally hidden and not visible to most users.

2.4.1. Special Case – Raw TAP, Wave, Atari, CBM, ZX80 / 81

Generic Recordings, Wave, Atari, Commodore & Sinclair ZX80 / 81 images are treated as a special case by **SVI-CAS**. The Atari and ZX80 machines do not support program file naming, while the ZX81 does, Commodore TAP and Wave images on the other hand are raw time-based digital copies of real analog recordings. Thus data for these situations is stored in images (regardless of format type) without any true easily accessible file header data – in addition to this, such images are often simply a single program and essentially in most cases an entire copy of the machine memory content. As a result, **SVI-CAS** will deal with images for these machines by using a generic entry representation for each program found as shown below:

Name : [Tape] Type : (Data)

Only **TZX** images for these Sinclair machines will potentially contain multiple programs that may be individually reachable by **SVI-CAS**, as the other supported formats contain no data indicating program size or starting point within the file, they are just either single raw copies of memory or digitized recordings of analog tape playback.

2.5. Play-Back Screen

The play-back screen is displayed when a user has made a prior selection and **SVI-CAS** has begun the process of converting the source image file back to an audio signal the host computer can recognize and load. During the play-back process **SVI-CAS** will display information concerning the play-back environment as well as a set of file specific attributes depending upon the image format and the host computer type.



Figure 2.5.1. Sample MSX Play-Back Display

The play-back operation when active has several logic states, these being:

- **READY** – preparations to begin playback have completed
- **WAITING** – waiting for the user to press the touch screen to continue
- **PLAYING** – currently playing leader tone or a block of data
- **PAUSE** – pausing for a specific amount of time (silence between blocks)

The “**WAITING**” logic state is only valid where play-back control has been set to manual and requires the user to press ▷ after every program section to continue with the play-back / loading process. Under special circumstances a **TSX** / **TZX** image file may contain a tape stop command which will result in play-back pausing regardless, the user having to press ▷ to continue.

Each program (each section as well) comprises a collection of blocks, the reproduction of some requiring the playing of a leader tone for a period of time, then immediately playing the accompanying data, finally finishing with a pause of silence. Where a block is the final block in an image the pause stage will be skipped and play-back will be considered to have completed. Note that the attached computer may decide to end loading early and leave the image permanently paused till the user intervenes to stop play-back manually – this can only occur where a machine has the ability to control tape flow (supported non-Sinclair machines) and that feature is connected (via a lead) to **SVI-CAS**.

The following sub-sections describe the play-back display content in more detail.

2.5.1. Heading

The heading content defines the expected host machine type, this will be either the SVI-318 / 328 computers, Tandy TRS-80 CoCo, MSX, Amstrad, Acorn, Commodore, or a Sinclair based machine.

2.5.2. Interface

Intface: (<baud>) [<control type> : <signal phase>]

The interface information indicates both the selected baud rate and the type of play-back control being exercised. If a **CAS** type image is being played back then the baud rate displayed will be the menu selection made by the user, if a **TSX** / **TZX** image is being played back the baud rate will be automatically derived from information stored within each block (auto), or shown as per the **CAS** option if the baud rate control is set to manual.

It should be noted that (unlike MSX & Acorn) baud rates for the SVI-318 / 328, TRS-80 CoCo, & Sinclair machines are fixed for images and are not user selectable.

Play-back control is specified as automatic “[A:]” or manual “[M:]” depending upon the main menu setting. The second attribute specifies the output signal as inverted or non-inverted depending upon the set output requirement (main menu). It should be noted that some images require play-back via a specific output setting else the host machine will not successfully load a program. Representation is specified as inverted “:I]” or normal “:N]” depending upon the main menu setting.

2.5.3. Format

Format: <source image type>

Specifies the type of source image being processed :

- **Atari 400 / 800 / XL / XE Series**
 - CAS – displayed as “CAS Image”
- **SVI-318 / 328**
 - CAS – displayed as “CAS Image”
 - TSX – displayed as “TSX Image”
- **TRS-80 CoCo / MC10**
 - CAS – displayed as “CAS Image”
- **Dragon 32 / 64**
 - CAS – displayed as “CAS Image”
- **MSX machines**
 - CAS – displayed as “CAS Image”
 - TSX – displayed as “TSX Image”
- **Sinclair ZX80**
 - ‘.O’ – displayed as “OZX Image”
 - TZX – displayed as “TZX Image”
- **Sinclair ZX81**
 - ‘.P’ – displayed as “PZX Image”
 - TZX – displayed as “TZX Image”
- **Sinclair ZX Spectrum**
 - TAP – displayed as “TAP Image”
 - TZX – displayed as “TZX Image”
- **Amstrad CPC machines**
 - CDT – displayed as “CDT Image”
- **Acorn machines**
 - UEF – displayed as “UEF Image”
- **Commodore machines**
 - TAP – displayed as “TAP Image”
- **Oric-1 / Atmos**
 - TAP – displayed as “TAP Image”
- **Wave Image**
 - WAV – displayed as “WAV Image”
- **SVI_CAS Recordings**
 - TAP – displayed as “TAP Image”

2.5.4. Block (Updates for CDT/TSX/TZX images)

Block: #<block number> <block type>

Specifies the number of the current block being played and also the block type, for **TSX / TZX** images only. When starting play-back the block number may not start at one, this is because either the selected entry is not at the start of an image, or the image contains additional blocks of data that are for alternative purposes and not required for play-back and are skipped in order to save time and help load the image as quickly as possible. Some blocks may be processed so quickly there is no actual time to display them before the next block is reached and must be dealt with.

The specific block types that **SVI-CAS** will inform the user of are as follows:

- **ID 10** – Standard speed data block
- **ID 11** – Turbo speed data block
- **ID 12** – Pure tone block
- **ID 13** – Pulse sequence block
- **ID 14** – Pure data block
- **ID 15** – Direct recording block
- **ID 19** – General record block
- **ID 20** – Pause block
- **ID 21** – Group start block
- **ID 22** – Group end block
- **ID 24** – Loop start block
- **ID 25** – Loop end block
- **ID 4B** – Kansas City Standard block

2.5.5. Chunk (Updates for UEF images)

Chunk: #<chunk number> <chunk type>

Specifies the number of the current chunk being played and also the chunk type, for **UEF** images only. When starting play-back the chunk number may not start at one, this is because either the selected entry is not at the start of an image, or the image contains additional chunks of data that are for alternative purposes and not required for play-back and are skipped in order to save time and help load the image as quickly as possible. Some chunks may be processed so quickly there is no actual time to display them before the next chunk is reached and must be dealt with.

The specific chunk types that **SVI-CAS** will inform the user of are as follows:

- **ID 100** – Implicit start/stop bit tape data chunk
- **ID 102** – Explicit tape data chunk
- **ID 104** – Defined tape format data chunk
- **ID 110** – Carrier tone
- **ID 111** – Carrier tone with dummy byte
- **ID 112** – Integer gap
- **ID 114** – Security cycles
- **ID 116** – Floating point gap

2.5.6. Chunk (Updates for Atari CAS images)

Chunk: #<chunk number> <chunk type>

Specifies the number of the current chunk being played and also the chunk type, for **Atari CAS** images only. When starting play-back the chunk number may not start at one, this is because either the selected entry is not at the start of an image, or the image contains additional chunks of data that are for alternative purposes and not required for play-back and are skipped in order to save time and help load the image as quickly as possible. Some chunks may be processed so quickly there is no actual time to display them before the next chunk is reached and must be dealt with.

Note that SVI-CAS implements a sub-set of the full Atari CAS format, implementing those chunks that are known to exist within the existing library of Atari cassette images.

The specific chunk types that **SVI-CAS** will inform the user of are as follows:

- **ID DATA** – Standard Atari SIO record
- **ID FSK** – Non-standard Atari SIO signals

2.5.7. Source

Source: <source image file name>

Specifies the file name of the source image, name will scroll if longer than 12 characters in length, 255 character length file names supported. Scrolling will stop after play-back has ended successfully or a failure has occurred.

2.5.8. Size

Size: <size in bytes of image source file>

Specifies the image source file size in bytes, this is the total file size and not that of any selected internal entry. The maximum file size that may be displayed is capped at 9,999,999 for bytes, larger files will be divided by 1000 and displayed in kilobytes.

2.5.9. File Name

Fname: <name of current entry being played back>

Specifies the internal SVI/TRS-80/MSX file contained within an image file currently being played back. This file name will be up to 6 characters long for SVI and MSX computers, while up to 8 characters for the TRS-80 CoCo, and 10 characters maximum for Amstrad, Acorn, and the ZX Spectrum. No file name is displayed for the Atari, ZX80, ZX81, & Commodore machines. File names may consist of any valid character including only spaces for the respective machine.

2.5.10. File Type

Ftype: <type of current entry being played back>

Specifies the internal SVI/TRS-80/MSX/Amstrad/Acorn/Sinclair file type contained within an image file currently being played back. It should be noted that the Acorn file standard provides no means of identifying the actual file type after reading the file header, in this particular situation the file type is simply displayed as 'Acorn'.

See section 2.4. for further information.

2.5.11. Current Play-Back State

Specifies the current status of the play-back process, “**READY**”, “**WAITING**”, “**PLAYING**”, and “**PAUSE**”. The “**WAITING**” state only occurs during manual play-back operations.

2.5.12. Progress Bar

Specifies the how much of the image has been processed / played-back so far, this is measured against the overall size of the source image file and not against individual image entry sizes. Result is displayed graphically as a percentage between 0 and 100.

The animation of the progress bar may be smooth from start to finish during a playback session, or depending upon the image type, appear to stall while generating pilot tone, then restart when actual data is being transmitted. Additionally, should the user elect to start playback from some position other than the absolute beginning, the progress bar will show the starting point as an offset of the total image length.

A typical sample play-back display is depicted as follows in **Figure 2.5.12.1.** below:



Figure 2.5.12.1. Play-back Screen

A play-back session may end in one of three ways, firstly it may complete successfully with a “**COMPLETED**” indication, or it may be ended manually by the user pressing the “STOP” button resulting in an “**ABORTED**” message, or finally the source image file may be faulty/flawed and result in an “**ERROR**” message being generated.

In addition to this, a play-back session may also make use of the tape memory facility which permits the recording of a particular starting point within the selected image that can be revisited at a later stage of image use multiple times – for example, the re-loading of “stage 1” level data for a game program. A new image starting point may be repeatedly set by the user as they see fit.

For the Sinclair ZX80 & ZX81 the tape memory feature is of no value for **OZX** and **PZX** image types due to these images being single memory images. Images in the **TZX** format for these machines often contain compilations of programs and will benefit from the use of this memory feature.

While playing, if manual play-back was selected from the main menu, **SVI-CAS** will pause and remain waiting for the user to press the ▷ button after each block regardless of the state of the tape motor control. It must be remembered that such a manual interruption may disrupt the current loading process and force the user to resort to starting from the beginning again though.

Table 2.5.12.1. which follows, describes the button-based operation of this particular screen while image play-back is in progress.

Function	Options
	Continue with current play-back, disabled if button text is RED in colour.
	Pause play-back operation, disabled if button text is RED in colour.
	Memorize the current play position, disabled if button text is RED in colour.
	Stop playing the current image.

Table 2.5.12.1. Active Play-Back Buttons

Table 2.5.12.2. which follows, describes the button-based operation of this particular screen after image play-back has either completed successfully, or the user has pressed the  button to end play-back (abort).

Function	Options
	Re-start image play-back at the last memorized play point.
	Re-start image play-back at the beginning of the image.
	Return to the image inspection menu.
	Go back one step or directory level.

Table 2.5.12.2. Stopped / Aborted / Completed Play-Back Buttons

2.6. Tape Memory Operation

The tape memory option is very easy to use once the operational behavior of this feature is understood by the user. It is particularly handy for games programs that ask the user to “Stop the Tape!” or to pause playback and reset the tape counter to zero after the initial loading period, allowing the user to mark and remember the start of game level data for future re-loading when a game has ended and is re-started.

When recording the tape memory position, **SVI-CAS** will attempt to remember the starting point of the current block or chunk for some image formats (**TZX**, **UEF**), else the starting point of the very next block or chunk will be recorded. In the instance the image being played has no discernable block or chunk style composition, the exact current byte level position will be recorded instead.

To use the memory feature the following actions are carried out:

1. Press the  button to pause play-back.
2. Press the  button to memorize the start of the next data block.
3. The user may now either press the  button to continue playing OR press the  button and play from the original or the memorized starting points using the  or  buttons.

In general, the  button will start playback from the original selected starting point, aka the beginning of the original playback operation and the  button will restart playback from the currently memorized position.

By default if no position is memorized the  button will have the same effect as the  button and start playback from the original selected starting point.

2.6.1. TSX & TZX Specifics

The best point at which to use this feature is to pause play-back while the pause message is being displayed – right after a program has finished loading. Some **TSX** & **TZX** images will (as previously mentioned) pause automatically and change the status of the play-back screen button automatically after a program is loaded – this feature is not present in all images, but appears mostly in a number of ZX Spectrum **TZX** game images, for example the game “*ATHENA*” by Imagine software is one such image that has this command added to it. It is suggested that the user try out such an image in order to acquaint themselves with this behavior.

Note: Pressing pause and memorizing the start of the next block just before one of the automatic pause blocks can result in a “double” pause where the user will have to press the ▷ button twice to continue, this is not a fault, the user has just elected to remember a play-back point which happens to be an automatic pause request for **SIV-CAS**!

A very important point to remember: Whenever the ○ button is pressed the starting point of the very next block is stored, **NOT** the position in the current block, that is why it is best to use this feature when play-back has reached a pause point where a block has finished playing and silence is being generated for a period of time. Pausing play-back during an actual block will generate a loading error on the host computer if data is in the process of being played, though the starting point of the next block will still be recorded.

In other words - **TSX** and **TZX** images may contain blocks that may be referred to as sub-blocks (i.e. “collections” of data) that make-up an entire leader tone – data – silence stage of a program and hence it is not useful to pause and record at that stage, this function is best used when silence (pause) is active, or when the image player has detected an embedded request to pause play-back.

2.6.2. CAS, CDT, UEF, & ZX TAP Specifics

Tape memory operation for these images formats are equivalent to that of **TSX** & **TZX** images, in that the start of the very next block is memorized, hence the prior documented general specifics apply in these cases as well.

2.6.3. Wave, Commodore TAP, ZX80/81 Specifics

Tape memory operation for Wave images, Commodore and ZX80/81 images are very specific, with the actual byte position within the image being recorded as the memorized starting point for these raw forms of image. There are no blocks to be skipped or jumped to, this is strictly the marking of a starting point for repeated raw data playback.

3. SVI-CAS HOST COMPUTER CONNECTIONS

Refer to the following diagrams for the correct connection of each supported computer brand & model described in the sections that follow. Please note that these machines have their own data port separate from all other machines.

3.1 General connection

The following diagram illustrates general connectivity for *SVI-CAS*:

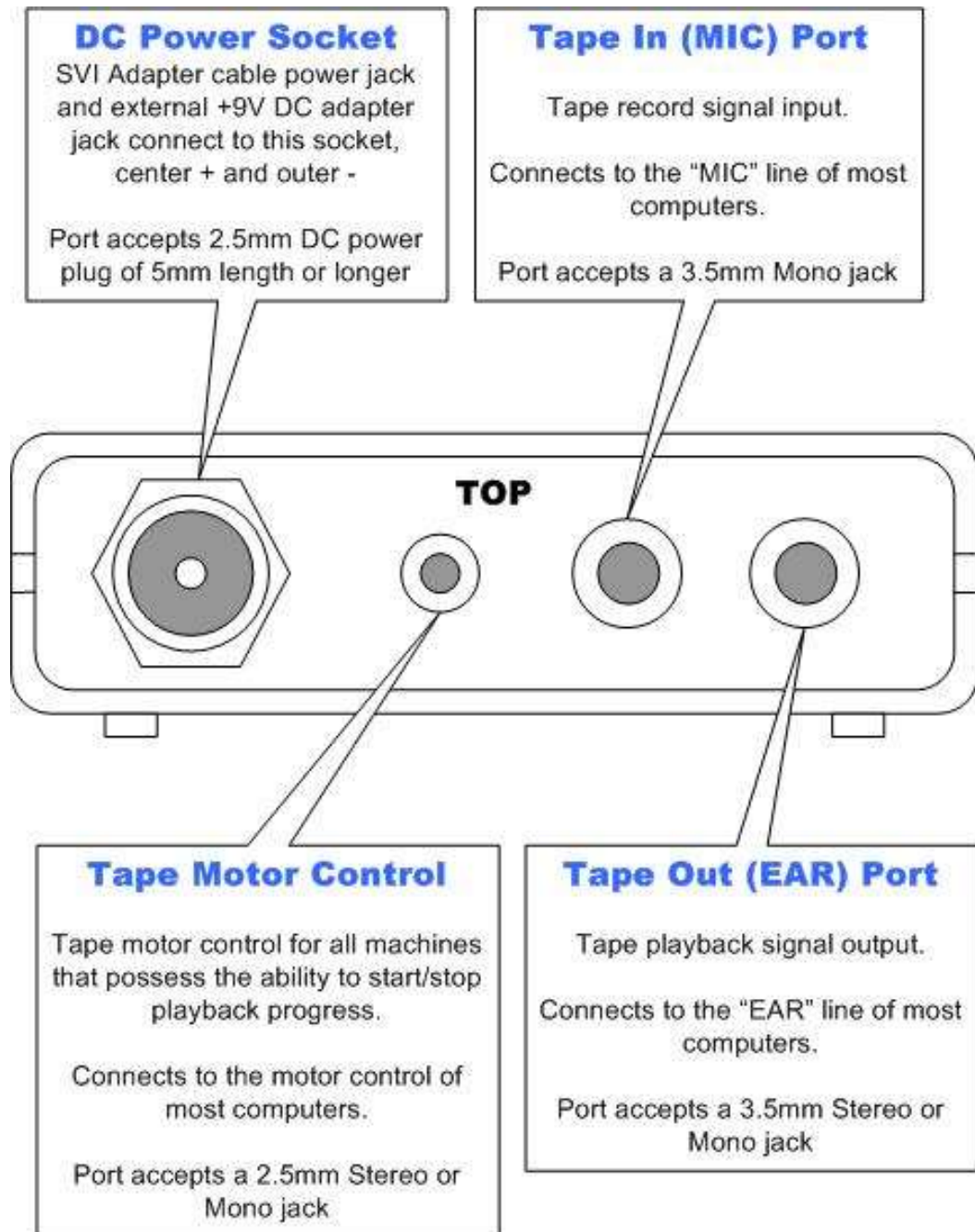
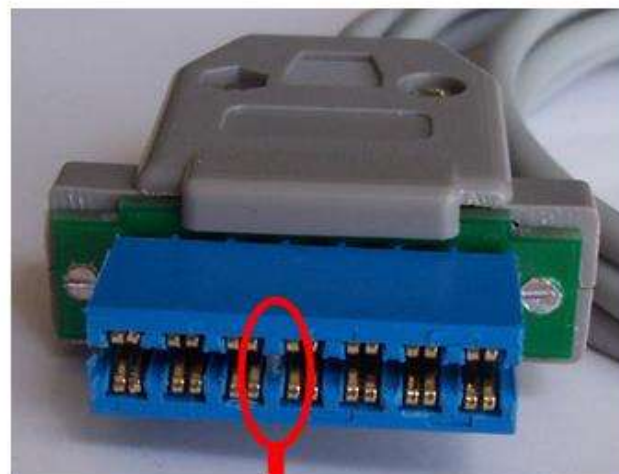
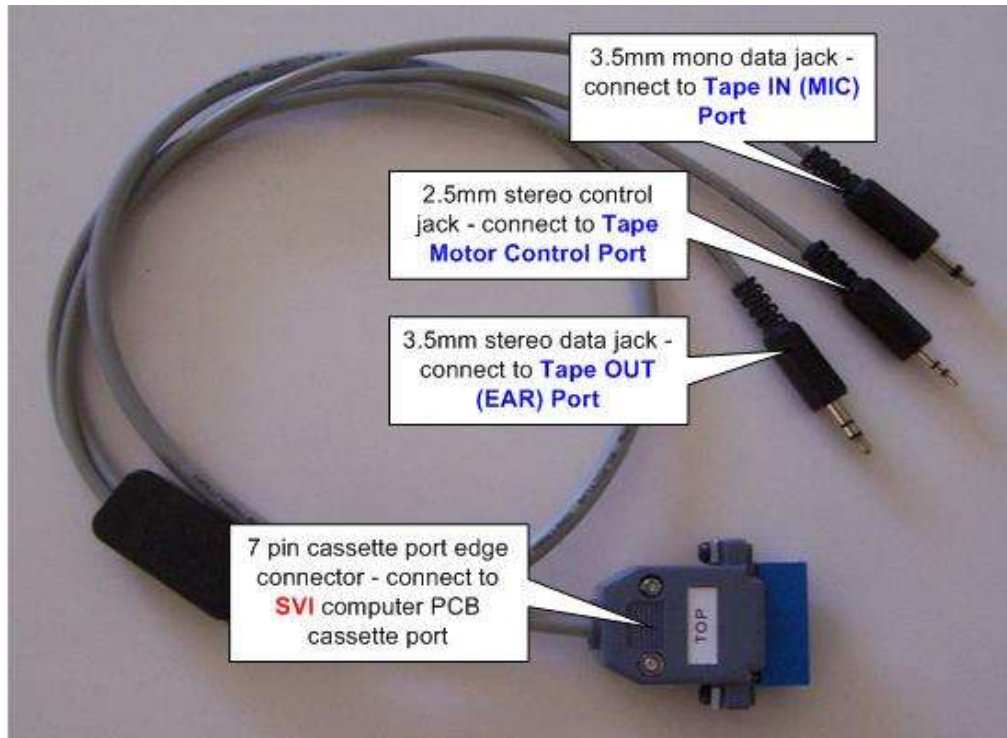
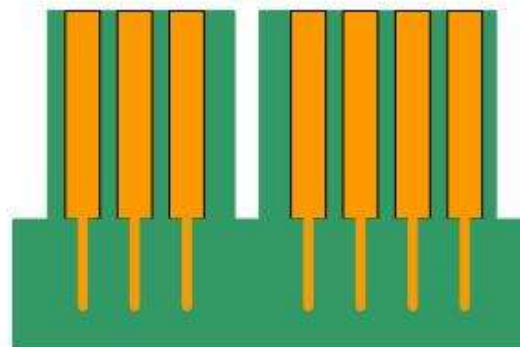


Figure 3.1.1. SVI-CAS connections.

3.2 Spectravideo Adapter Cable



KEY - Correct alignment is critical !!!



SVI PCB Cassette Port
Top View

Figure 3.2.1. SVI-318/328 inter-connections.

3.3 Commodore Adapter Cable

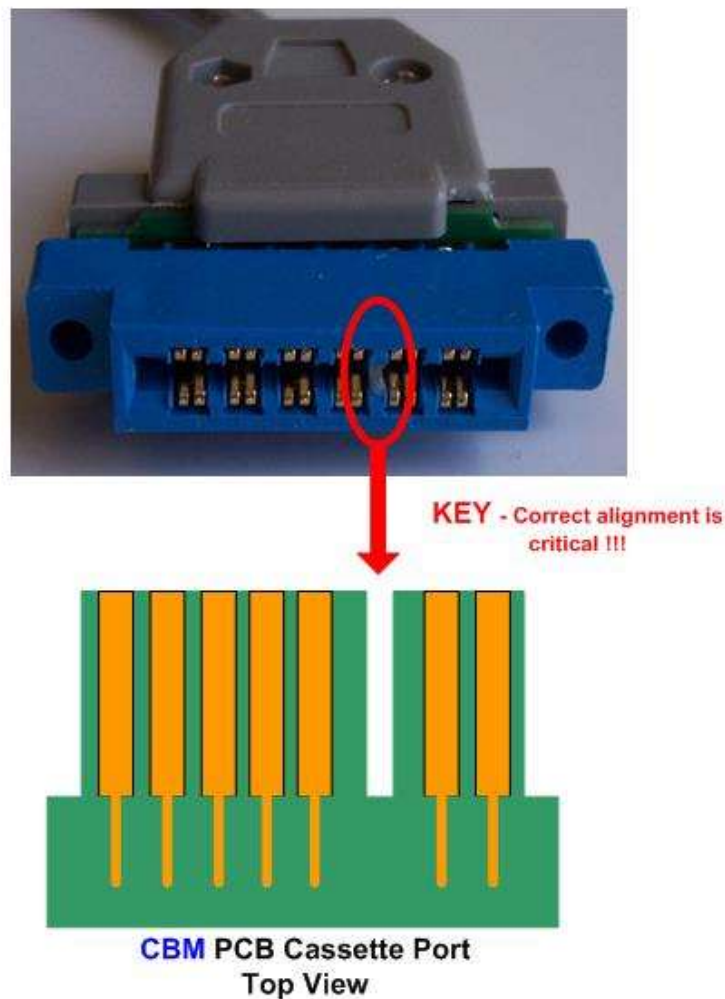
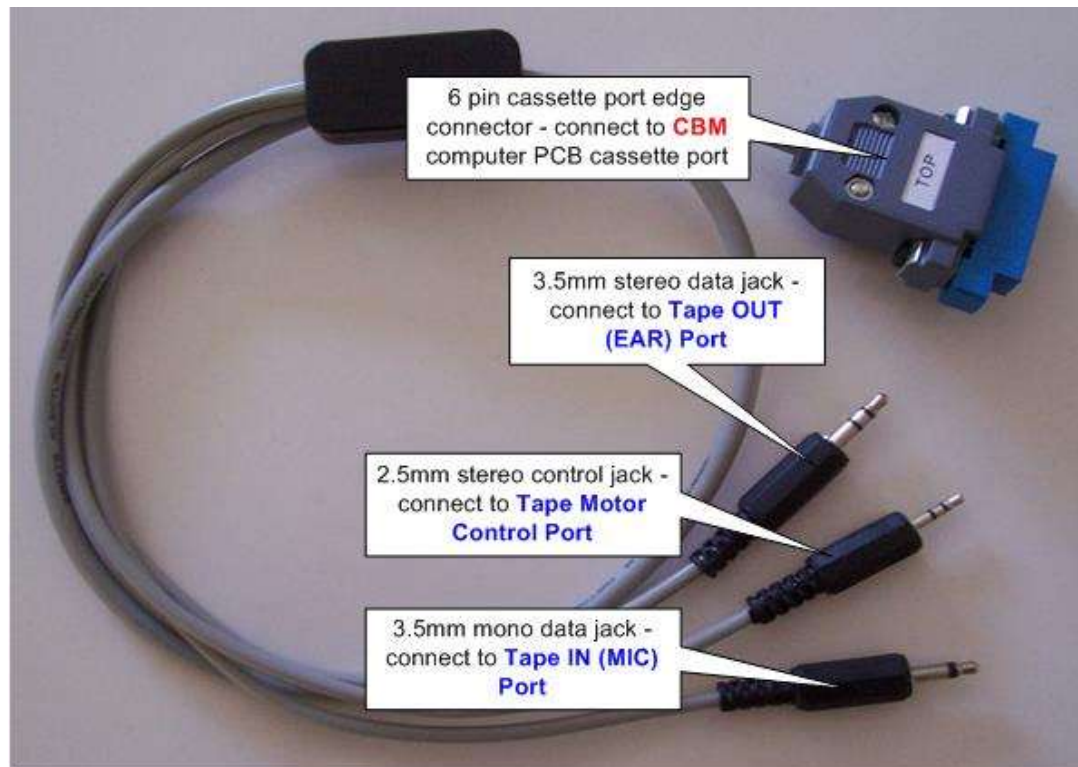


Figure 3.3.1. Commodore computer inter-connections.

3.4 Atari Adapter Cable

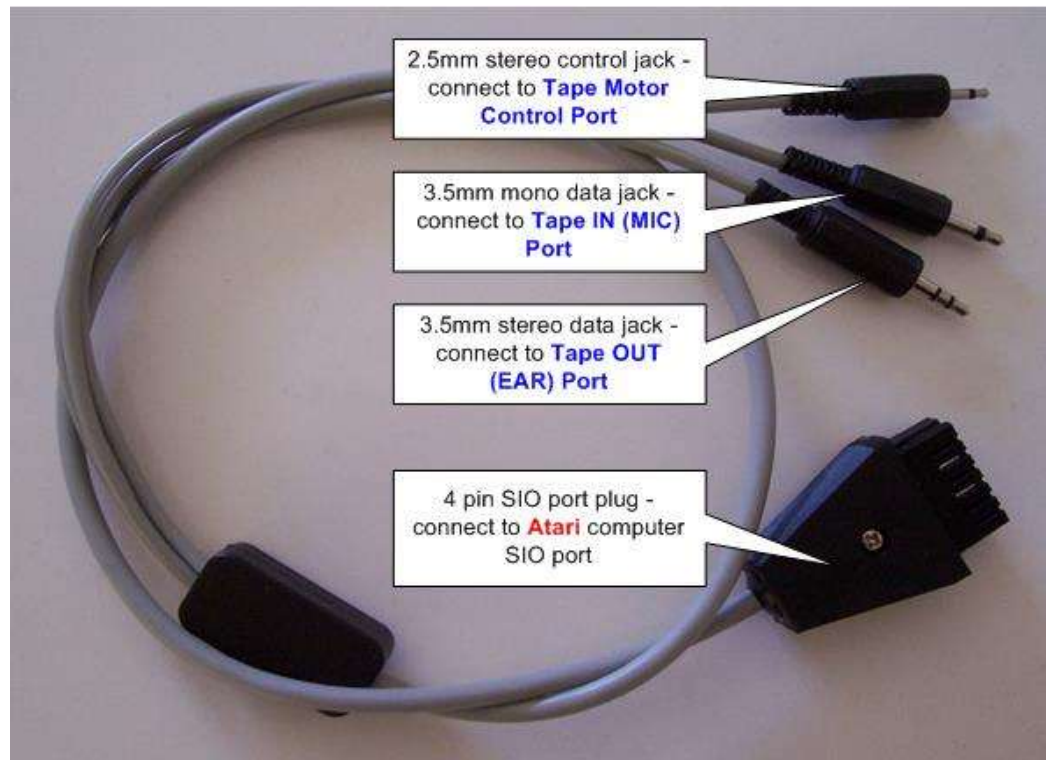


Figure 3.4.1. Atari computer inter-connections.

3.5 Commodore 16 Additional Requirement

The Commodore 16 unlike earlier 8-bit Commodore machines, replaced the cassette edge connector method with a MINI-DIN connector instead, as a result the complexity of attaching an **SVI-CAS** to a Commodore 16 is higher than with other machines of the family. The user will need to acquire an additional adapter cable that provides replication of the older original PCB fingers to attach to the **SVI-CAS** adapter cable, while providing a suitable MIN-DIN connector at the other end of the cable.

The author of **SVI-CAS** does not manufacture such a cable, but they can be bought from sellers on EBay – perform a search for auctions concerning a Arduino device called “TAPUINO” and examine the seller’s other items, in that list the required adapter cable may be found.



Figure 3.4.1. Examples of C16 adapter cables from EBay.

4. SVI-318/328 PLAYBACK

This section describes the general characteristics and operational specifics of **SVI-CAS** when used with the Spectravideo SV-318 / 328 computers.



Figure 4.1. SVI-318 / 328 Computer Operation

The SVI-318/328 computers differ from other supported machines in that they require use of a specific Spectravideo built and supplied cassette unit, accordingly the relationship between computer and cassette unit is more significant in that the sensing of cassette unit key states becomes a critical aspects of cassette based operations. Thus a slightly different approach must be taken when compared to other **SVI-CAS** supported computers when loading images as is described in the following steps. By default, the cassette motor is always enabled by tape drive key presses to allow tapes to be re-wound or fast-forwarded – which creates an issue when starting image play-back as will be dealt with in the steps that follow.

The following special adapter cable is required for the SVI-318/328 machines:



Figure 4.2. SVI Adapter Cable



Figure 4.3. SVI-318/328 Connection

1. Connect **SVI-CAS** to the computer cassette port, ensuring that the correct orientation of the **SVI-CAS** plug is maintained and that the plug is correctly aligned via the plug key and cassette port key-gap of the protruding PCB contact fingers. In addition to this key the top of the plug is marked with a blue label to indicate it should face upwards.

NEVER twist the cassette port plug to insert or remove it, this may result in damage or even dislodgement of the plug key and can cause serious electrical damage to both **SVI-CAS** and the host computer should adjacent contacts be shorted together.

Connect the 3.5mm Stereo jack to the **Tape OUT** port, the 3.5mm Mono jack to the **Tape IN** port, 2.5mm jack to the **Tape Motor Control** port and insert the AC adapter power jack into the power socket. If unsure as to which sockets to use please consult section 3 before continuing.

NOTE: Attempting to connect **SVI-CAS** using an alternative means may result in damage to **SVI-CAS**, your computer, or both, this adapter cable is specially made for use with this image player and is more than simply a collection of leads & connectors.

If the user lacks (or has lost) this cable they should contact the author to arrange purchase via the same means with which **SVI-CAS** was originally obtained. Always check the plug key after removing the cable to ensure it is present and intact.

2. Power up the computer and **SVI-CAS**, ensuring that the computer boots up correctly into BASIC and that **SVI-CAS** displays a splash screen followed by the main menu. If the computer and/or **SVI-CAS** fail to power up, immediately turn off the computer and check all connections.
3. Using **SVI-CAS** ensure that the correct interface is selected (SVI-318 / 328) and that "Automatic Play" is enabled.
4. Move to the directory menu and select the desired image to be loaded, be sure to select an image that is actually intended for this computer, Press  to move to the image inspection menu.
5. On the computer enter the required command to start the loading process, the command used will depend on the selected image entry, see below:
 - For **(Basic)** files use **CLOAD <ENTER>**
 - For **(Ascii)** files use **RUN"CAS:" <ENTER>**
 - For **(Binary)** files use **BLOAD"CAS:",R <ENTER>**

After entering the required command the computer will ask the user to press the play button on the cassette unit.

6. Select the desired entry inside the image which you would like loading to commence with and press  or  to start the playback process. **SVI-CAS** will now wait for the computer to assert the cassette motor control line before it will actually start generating audio.
7. Upon detection of the tape motor line being asserted the play-back process should begin. Watch the LCD display and check that the name of your selected entry appears after a short time, also verify that the computer also displays this same entry name as well during loading.
8. Throughout the loading process **SVI-CAS** will cycle between the "**PLAYING**" and "**PAUSE**" stages until the desired program has finished loading. If the loading process starts with a (Basic) file it will be necessary for the user to manually "**RUN**" this program in order for any further loading to proceed, as **SVI-CAS** will remain paused while the computer de-asserts the cassette motor signal after the initial **CLOAD** operation. Type the command or press "F3" on the Spectravideo keyboard to continue loading.
9. When all required loading is completed and the program is up and running on the computer examine the state of **SVI-CAS**. If **SVI-CAS** is paused then this is usually because a program was loaded that was part of a compilation. **SVI-CAS** cannot determine where a complete program starts or ends.... It will continue to play further blocks of data till the overall end of an image file is reached if asked to do so. Hence if **SVI-CAS** participation has ended then the user should exit back to the main menu.
10. It is suggested that should a loaded program be used for an extended period of time the user might consider removing the power lead from **SVI-CAS** to reduce loading of the host SVI computer's power supply.

****Note:** Some SVI software (e.g. the SVI game Spectron) do not bother to disable cassette operation after the program has finished loading – this means a real cassette unit would continue playing the inserted cassette till the end of the tape was reached. Like-wise if the computer does not bother to de-assert the cassette motor control line that **SVI-CAS** relies on to switch between play and pause activities, then it will continue to play the rest of the image until the end of file is encountered. In this case the user must manually instruct **SVI-CAS** to stop playing via pausing or ending play-back.

5. CoCo, MC10 & DRAGON 32 / 64 PLAYBACK

This section describes the general characteristics and operational specifics of **SVI-CAS** when used with the Tandy TRS-80 Color Computer, MC10 or Dragon Data Dragon 32 & 64 computers.



Figure 5.1. TRS-80 Color Computer & Dragon 32 / 64 Operation

For simplicity only the Color Computer will be directly discussed, as the MC10 and Dragon computers use the exact same commands and behave in the exact same manner when loading programs, hence follow the same procedures when using an MC10 or Dragon versus a Color Computer.

The TRS-80 CoCo like the majority of early 1980s computers makes use of an external generic cassette unit for the loading of programs. To achieve best results it is recommended that the user makes use of a lead that not only connects audio to the computer, but also allows for the computer to control the playing and pausing of an image via tape motor control, some software requires this to load successfully.

If such a lead is not available it is still possible to connect only audio out from **SVI-CAS** to the computer, though the user will have to manually start the play-back of each image block by pressing ▷ when prompted by **SVI-CAS** through its **“WAITING”** message.

1. Connect **SVI-CAS** to the computer cassette port, ensuring that the 3.5mm “EAR” jack of the lead is plugged into the **Tape OUT** port, the 3.5mm “MIC” jack of the lead is plugged into the **Tape IN** port, while also ensuring the 2.5mm jack is connected to the **Tape Motor Control** port. The DIN connector of the lead is plugged into the TRS-80 Color Computer cassette port. If unsure as to which sockets to use consult section 3.
2. Plug in the AC adapter and power-up **SVI-CAS**, ensuring it defaults to the main menu, then power-up the computer and ensure it boots correctly into BASIC.
3. Using **SVI-CAS** ensure that the correct interface is selected as **“TRS-80 CoCo”** and that **“Automatic Play”** is enabled if the connecting lead supports it, else change to **“Manual Play”**.



Figure 5.2. Basic connections

4. Move to the directory menu and select the desired image to be loaded, be sure to select an image that is actually intended for this computer! Press  to move to the image inspection menu.
5. Select the desired entry inside the image which you would like to load, then press the  or  button. **SVI-CAS** will move to the play-back screen and display “**READY**” if it is waiting for the computer to initiate play-back, else it will display “**WAITING**” if the user is required to manually start play-back of the next image block.
6. On the computer enter the required command to start the loading process, the command used will depend on the selected image entry, see below:
 - For **(Basic)** files use **CLOAD <ENTER>**
 - For **(Binary)** files use **CLOADM <ENTER>**
7. After starting the loading process on the computer press the  button if loading manually when prompted to do so, else wait for the computer to finish loading the program. Watch the LCD display and check that the name of your selected entry appears after a short time, also verify that the computer also displays this same entry name as well during loading.

8. Throughout the loading process **SVI-CAS** will cycle between the “**PLAYING**” and “**PAUSE**” stages until the desired program has finished loading. If the loading process starts with a (Basic) file it may be necessary for the user to manually run this program in order for any further potential loading to proceed, as **SVI-CAS** will remain paused when the computer de-asserts the cassette motor signal.
9. After completion of loading a BASIC program the user should enter a “**RUN**” command to start the program, else if the loaded program was a (Binary) file the user must type “**EXEC**” to start the loaded program.
10. When all required loading is completed and the program is up and running on the computer examine the state of **SVI-CAS**. If **SVI-CAS** is paused then this is usually because a program was loaded that was part of a compilation. **SVI-CAS** cannot determine where a complete program starts or ends.... It will continue to play further blocks of data till the overall end of an image file is reached if asked to do so. Hence if **SVI-CAS** participation has ended, the user should exit back to the main menu.

6. MSX PLAYBACK

This section describes the general characteristics and operational specifics of **SVI-CAS** when used with an MSX computer.



Figure 6.1. MSX Computer Operation

MSX computers like the majority of early 1980s computers make use of an external generic cassette unit for the loading of programs. To achieve best results it is recommended that the user makes use of a lead that not only connects audio to the computer, but also allows for the computer to control the playing and pausing of an image via tape motor control.

If such a lead is not available it is still possible to connect only audio out from **SVI-CAS** to the computer, though the user will have to manually start the play-back of each image block by pressing “**RESUME**” when prompted by **SVI-CAS** through its “**WAITING**” message.

1. Connect **SVI-CAS** to the computer cassette port, ensuring that the 3.5mm “EAR” jack of the lead is plugged into the **Tape OUT** port, the 3.5mm “MIC” jack of the lead is plugged into the **Tape IN** port, while also ensuring the 2.5mm jack is connected to the **Tape Motor Control** port. The DIN connector of the lead is plugged into the MSX computer cassette port. If unsure as to which sockets to use consult section 3.



Figure 6.2. Basic connections (will vary according to MSX computer)

2. Plug in the AC adapter and power-up **SVI-CAS**, ensuring it defaults to the main menu, then power-up the computer and ensure it boots correctly into BASIC.
3. Using **SVI-CAS** ensure that the correct interface is selected from one of several “**MSX <baud> Baud**” options and that “**Automatic Play**” is enabled if the connecting lead supports it, else change to “**Manual Play**”.
4. Move to the directory menu and select the desired image to be loaded, be sure to select an image that is actually intended for this computer! Press  to move to the image inspection menu.
5. Select the desired entry inside the image which you would like to load, then press the  or  button. **SVI-CAS** will move to the play-back screen and display “**READY**” if it is waiting for the computer to initiate play-back, else it will display “**WAITING**” if the user is required to manually start play-back of the next image block.

6. On the computer enter the required command to start the loading process, the command used will depend on the selected image entry, see below:
 - For **(Basic)** files use **CLOAD <ENTER>**
 - For **(Ascii)** files use **RUN"CAS:" <ENTER>**
 - For **(Binary)** files use **BLOAD"CAS:",R <ENTER>**
7. After starting the loading process on the computer press the  button if loading manually when prompted to do so, else wait for the computer to finish loading the program. Watch the LCD display and check that the name of your selected entry appears after a short time, also verify that the computer also displays this same entry name as well during loading.
8. When all required loading is completed and the program is up and running on the computer examine the state of **SVI-CAS**. If **SVI-CAS** is paused then this is usually because a program was loaded that was part of a compilation. **SVI-CAS** cannot determine where a complete program starts or ends.... It will continue to play further blocks of data till the overall end of an image file is reached if asked to do so. Hence if **SVI-CAS** participation has ended then the user should exit back to the main menu.

7. SINCLAIR PLAYBACK

This section describes the general characteristics and operational specifics of **SVI-CAS** when used with various Sinclair computers (ZX80, ZX81, ZX Spectrum).



Figure 7.1. Sinclair Computer Operation

Sinclair computers like the majority of early 1980s computers make use of an external generic cassette unit for the loading of programs. Unlike most machines of the era, Sinclair machines such as the ZX80, ZX81, and ZX Spectrum did not provide a means of tape motor control, requiring the user to manually manipulate tape play-back to meet the needs of the program being loaded.

SVI-CAS can improve this situation some-what, but still requires the user to be vigilant in respect of tape pausing when needed, it is here also that the tape memory function will prove valuable.

Audio out from **SVI-CAS** is connected to the “ear input socket” of the host computer and the user will have to monitor the play-back of an image and follow any instructions the host computer may issue on-screen, e.g. “Stop Tape NOW” for example.

1. Connect **SVI-CAS** to the computer cassette port, ensuring that the 3.5mm “EAR” jack lead is plugged into the **Tape OUT** port of the Spectrum & **SVI-CAS**, the 3.5mm “MIC” jack lead is plugged into the **Tape IN** port of the Spectrum & **SVI-CAS**, If unsure as to which sockets to use consult section 3.



Figure 7.2 Basic connections for ZX80, ZX81, ZX Spectrum

2. Plug in the AC adapter and power-up **SVI-CAS**, ensuring it defaults to the main menu, then power-up the computer and ensure it boots correctly into BASIC.
3. Using **SVI-CAS** ensure that the correct interface is selected for either the ZX80, ZX81, or ZX Spectrum depending upon the image to be loaded and that “**Automatic Play**” is enabled if using the ZX Spectrum with **TZX** images, as many ZX Spectrum software titles have explicit timing requirements as part of their tape copy protection schemes. Unlike **TZX** images, **TAP** images offer more timing flexibility and hence may be loaded with “**Manual Play**” if the user want play to halt at the end of each program section. In regards to the ZX80 & ZX81, it is suggested that “**Manual Play**” is used.
4. Move to the directory menu and select the desired image to be loaded, be sure to select an image that is actually intended for this computer! Press  to move to the image inspection menu.

5. Select the desired entry inside the image which you would like to load, but do not press  or  yet if using the free play option, as this will start playback before the host machine is ready since there is no tape motor control facility with the ZX machines.
6. On the computer enter the required command to start the loading process, the command used will depend on the selected image entry, see below:
 - For **ZX80** files use **LOAD <ENTER>**
 - For **ZX81** files use **LOAD"" <ENTER>**
 - For **ZX Spectrum** files use **LOAD"" <ENTER>**
7. After starting the loading process on the computer press the  or  to begin loading, pressing  again if manual tape control is selected (“**WAITING**” is displayed in manual mode when **SVI-CAS** is waiting for a user response) when prompted to do so, else wait for the computer to finish loading the program. Watch the LCD display and check that the name of your selected entry appears after a short time, also verify that the computer also displays this same entry name as well during loading.
8. Throughout the loading process **SVI-CAS** will cycle between the “**PLAYING**” and “**PAUSE**” stages until the desired program has finished loading. If the loading process was for a ZX80 or ZX81 it might be necessary to enter the RUN command to get the program started, in the case of the ZX Spectrum this is usually not required, 99% of commercial programs starting automatically after loading completes.
9. As has been previously discussed, should the program display a stop tape message it would be prudent to both pause play-back manually and then record the current tape position for future re-starting. See section **2.5.11.** for more information.
10. Some ZX Spectrum games are spread over multiple images, if prompted to turn over the tape and play side 2, just go back to the directory menu, select the next tape image for this program and then start playing it right from the start.

8. AMSTRAD CPC PLAYBACK

This section describes the general characteristics and operational specifics of **SVI-CAS** when used with an Amstrad CPC computer.



Figure 8.1. Amstrad Computer Operation

Amstrad CPC disk drive equipped computers like the majority of mid 1980s computers make use of an external generic cassette unit for the loading of programs. To achieve best results it is recommended that the user makes use of a lead that not only connects audio to the computer, but also allows for the computer to control the playing and pausing of an image via tape motor control.

For the Amstrad CPC 464 computer it will be necessary for the user to either use a cassette adapter that look like an actual cassette and is simply inserted into the built-in cassette unit or, install a 5 pin DIN connector conforming to the required pin connections of the disk based machines. It is suggested the user search the web for information on fitting such a connector, such information exists in multiple forms.

If the user opts for the cassette adapter instead of a fitted connector they will have to manually start the play-back of each image block by pressing “**RESUME**” when prompted by **SVI-CAS** through its “**WAITING**” message.

1. Connect **SVI-CAS** to the computer cassette port, ensuring that the 3.5mm “**EAR**” jack of the lead is plugged into the **Tape OUT** port, the 3.5mm “**MIC**” jack of the lead is plugged into the **Tape IN** port , while also ensuring the 2.5mm jack is connected to the **Tape Motor Control** port. The DIN connector of the lead is plugged into the Amstrad cassette port. If unsure as to which sockets to use consult section 3.

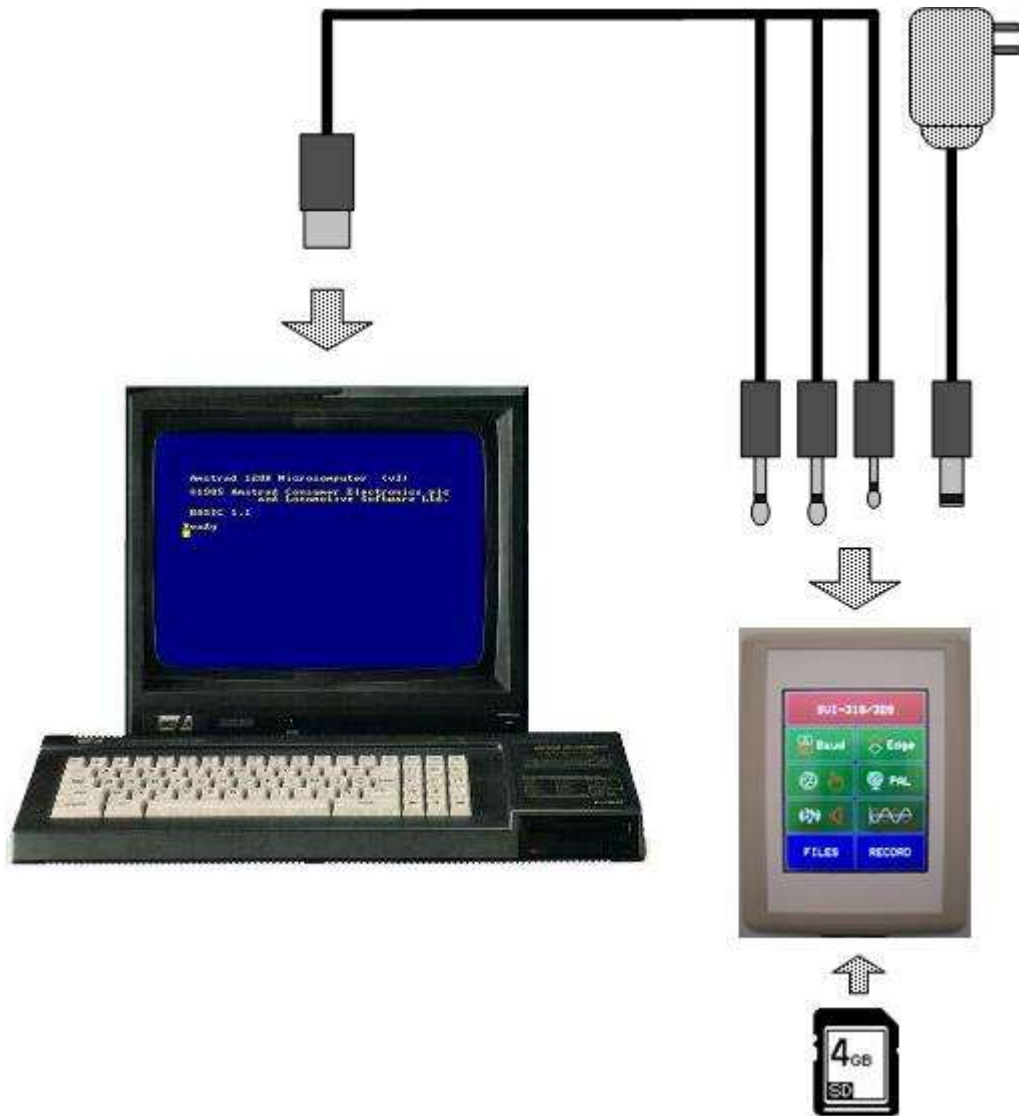


Figure 8.2. Basic connections

2. Plug in the AC adapter and power-up **SVI-CAS**, ensuring it defaults to the main menu, then power-up the computer and ensure it boots correctly into BASIC.
3. Using **SVI-CAS** ensure that the correct interface is selected i.e. “**Amstrad CPC**” and that “**Automatic Play**” is enabled if the connecting lead supports it, else change to “**Manual Play**”.
4. Move to the directory menu and select the desired image to be loaded, be sure to select an image that is actually intended for this computer! Press  to move to the image inspection menu.
5. Select the desired entry inside the image which you would like to load, then press the  or  button. **SVI-CAS** will move to the play-back screen and display “**READY**” if it is waiting for the computer to initiate play-back, else it will display “**WAITING**” if the user is required to manually start play-back of the next image block.

6. On the computer enter the required command to start the loading process, the commands used in the case of Amstrad CPC machines is always the same, first type “**|tape**” followed by the return key, then type “**run**” followed by the return key. If done correctly the computer will now ask you to press any key to start loading the program.
7. After starting the loading process on the computer press the  button if loading manually when prompted to do so, else wait for the computer to finish loading the program. Watch the LCD display and check that the name of your selected entry appears after a short time, also verify that the computer also displays this same entry name as well during loading.
8. When all required loading is completed and the program is up and running on the computer examine the state of **SVI-CAS**. If **SVI-CAS** is paused then this is usually because a program was loaded that was part of a compilation. **SVI-CAS** cannot determine where a complete program starts or ends.... It will continue to play further blocks of data till the overall end of an image file is reached if asked to do so. Hence if **SVI-CAS** participation has ended then the user should exit back to the main menu.

9. ACORN PLAYBACK

This section describes the general characteristics and operational specifics of **SVI-CAS** when used with an Acorn BBC Micro or Electron.



Figure 9.1. Acorn Computer Operation

Acorn computers like the majority of mid 1980s computers make use of an external generic cassette unit for the loading of programs. To achieve best results it is recommended that the user makes use of a lead that not only connects audio to the computer, but also allows for the computer to control the playing and pausing of an image via tape motor control.

If such a lead is not available it is still possible to connect only audio out from **SVI-CAS** to the computer, though the user will have to manually start the play-back of each image block by pressing “**RESUME**” when prompted by **SVI-CAS** through its “**WAITING**” message. Acorn image playback also requires that the playback phase of the image be set to 180 degrees instead of the default 0 degrees for all other machines, so be sure to press the phase button on the main menu, setting the highlighted phase to look like:

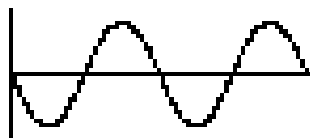


Figure 9.2. Required playback phase setting.

1. Connect **SVI-CAS** to the computer cassette port, ensuring that the 3.5mm “EAR” jack of the lead is plugged into the **Tape OUT** port, the 3.5mm “MIC” jack of the lead is plugged into the **Tape IN** port, while also ensuring the 2.5mm jack is connected to the **Tape Motor Control** port. The DIN connector of the lead is plugged into the Acorn computer cassette port. If unsure as to which sockets to use consult section 3.



Figure 9.3. Basic connections

9. Plug in the AC adapter and power-up **SVI-CAS**, ensuring it defaults to the main menu, then power-up the computer and ensure it boots correctly into BASIC.
10. Using **SVI-CAS** ensure that the correct interface is selected i.e. “**Acorn 1200 Baud**” and that “**Automatic Play**” is enabled if the connecting lead supports it, else change to “**Manual Play**”.
11. Move to the directory menu and select the desired image to be loaded, be sure to select an image that is actually intended for this computer! Press  to move to the image inspection menu.

12. Select the desired entry inside the image which you would like to load, then press the  or  button. **SVI-CAS** will move to the play-back screen and display “**READY**” if it is waiting for the computer to initiate play-back, else it will display “**WAITING**” if the user is required to manually start play-back of the next image block.
13. On the computer enter the required command to start the loading process, the commands used in the case of Amstrad CPC machines is always the same, first type “***TAPE**” followed by the return key, then type “**CHAIN**” for images where the first program encountered is a BASIC file, or “***RUN**” if it is a binary file followed by the return key. If done correctly the computer will now advise that it is searching for a program. In the event that loading went to completion but the program failed to run try using the opposite command in case the program was a binary file rather than a BASIC file and vice versa.
14. After starting the loading process on the computer press the  button if loading manually when prompted to do so, else wait for the computer to finish loading the program. Watch the LCD display and check that the name of your selected entry appears after a short time, also verify that the computer also displays this same entry name as well during loading.
15. When all required loading is completed and the program is up and running on the computer examine the state of **SVI-CAS**. If **SVI-CAS** is paused then this is usually because a program was loaded that was part of a compilation. **SVI-CAS** cannot determine where a complete program starts or ends.... It will continue to play further blocks of data till the overall end of an image file is reached if asked to do so. Hence if **SVI-CAS** participation has ended then the user should exit back to the main menu.
16. **SVI-CAS** offers the standard Acorn baud rate of 1200 as well as two higher baud rates of 1400 and 1500, these higher rates are non-standard and may not work with all Acorn machines. It is suggested that the user first try 1400 with their machine and should that prove successful then try 1500 for faster loading times.

NOTE: UEF images for the Acorn machines normally come zipped when downloaded from internet archive sites even though their suffix is “UEF” and the user will need to unzip images in order to use them with this image player. If **SVI-CAS** will not play a UEF image or complain that it is not supported when trying to start playback, then it is likely that the image is still zipped if it is in fact a genuine UEF file.

10. COMMODORE PLAYBACK

This section describes the general characteristics and operational specifics of **SVI-CAS** when used with a Commodore computer such as the VIC-20, Commodore 16, Commodore 64, or Commodore 128.



Figure 10.1. Commodore Computer Operation

Computer computers differ from other supported machines in that they require use of a specific Commodore built and supplied cassette unit, accordingly the relationship between computer and cassette unit is more significant in that the sensing of cassette unit key states and the sourcing of power from the host computer become critical aspects of cassette based operations. Thus a slightly different approach must be taken when compared to other **SVI-CAS** supported computers when loading images as is described in the following steps. By default, the cassette motor is always enabled by tape drive key presses to allow tapes to be re-wound or fast-forwarded – which creates an issue when starting image play-back as will be dealt with in the steps that follow.

The following special adapter cable is required for Commodore machines:



Figure 10.2. Commodore Adapter Cable

For Commodore loading operations it is important to ensure that two specific settings are correct, these are accessed via the main menu of **SVI-CAS**. The settings are the video format of the host computer (**NTSC** or **PAL**) and that the playback phase is 0 degrees, the first is determined by the machine type – North American & Japanese builds of Commodore machines and also some South-American units are all **NTSC**, while the rest of the world uses the **PAL** video standard, the user should thus select the correct video setting for their host machine. Then the user should ensure that the playback phase is set to 0 degrees, the correct setting will show the following waveform highlighted in white.

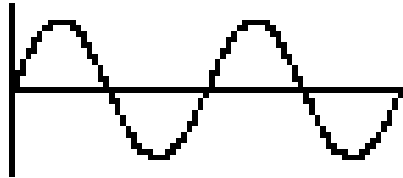


Figure 10.3. Required playback phase setting.



Figure 10.4. Commodore Connection

1. Connect **SVI-CAS** to the computer cassette port ensuring that the correct orientation of the **SVI-CAS** plug is maintained and that the plug is correctly aligned via the plug key and cassette port key-gap of the protruding PCB contact fingers. In addition to this key the top of the plug is marked with a blue label to indicate it should face upwards.

NEVER twist the cassette port plug to insert or remove it, this may result in damage or even dislodgement of the plug key and can cause serious electrical damage to both **SVI-CAS** and the host computer should adjacent contacts be shorted together.

Connect the 3.5mm Stereo jack to the **Tape OUT** port, the 3.5mm Mono jack to the **Tape IN** port, 2.5mm jack to the **Tape Motor Control** port and insert the AC adapter power jack into the power socket. If unsure as to which sockets to use please consult section 3 before continuing.

NOTE: Attempting to connect **SVI-CAS** using an alternative means may result in damage to **SVI-CAS**, your computer, or both, this adapter cable is specially made for use with this image player and is more than simply a collection of leads & connectors.

If the user lacks (or has lost) this cable they should contact the author to arrange purchase via the same means with which **SVI-CAS** was originally obtained. Always check the plug key after removing the cable to ensure it is present and intact.

2. Power up the computer, **SVI-CAS** should power-up at the same time, ensure that the computer boots up correctly into BASIC and that **SVI-CAS** displays a splash screen followed by the main menu. If the computer and/or **SVI-CAS** fail to power up, immediately turn off the computer and check all connections.
3. Using **SVI-CAS** ensure that the correct interface is selected (Commodore Model) and that "Automatic Play" is enabled.
4. Move to the directory menu and select the desired image to be loaded, be sure to select an image that is actually intended for this computer, Press  to move to the image inspection menu.
5. On the computer enter the required command to start the loading process, the command is invoked by holding down the left **SHIFT** key and then pressing and releasing the **RUN/STOP** key at the same time, then releasing the **SHIFT** key after releasing the **RUN/STOP** key. The host computer will respond with "**PRESS PLAY ON TAPE**"
6. Select the desired entry inside the image which you would like loading to commence with and press  or  to start the playback process. **SVI-CAS** will now wait for the computer to assert the cassette motor control line before it will actually start generating audio.
7. After **SVI-CAS** detects the computer asserting the tape motor line play-back the process should begin. Watch the LCD display and check that the name of your selected entry appears after a short time, also verify that the computer also displays this same entry name as well during loading. In the case of the VIC-20, the computer will display "**SEARCHING**" followed by "**FOUND <file name>**" when **SVI-CAS** has played enough data. The Commodore 64 & 128 behave slightly differently – their screens will go blank and upon finding the start of a program, briefly return to display the found program name.

8. Throughout the loading process **SVI-CAS** will cycle between the “**PLAYING**” and “**PAUSE**” stages until the desired program has finished loading. It should be noted that much of the commercial software developed for Commodore machines use custom loaders which will behave in ways that are different to standard Commodore loading behavior. After loading has completed the host Commodore computer will automatically start running the chosen program.
9. When all required loading is completed and the program is up and running on the host computer examine the state of **SVI-CAS**. If **SVI-CAS** is paused then this is usually because a program was loaded that was part of a compilation or there is extra sampled/unused data at the end of the image file. **SVI-CAS** cannot determine where a complete program starts or ends.... It will continue to play further blocks of data till the overall end of an image file is reached if asked to do so. Hence if **SVI-CAS** participation has ended then the user should exit back to the main menu on the provision that the program will not want to load this additional data in the near future, if in doubt just use the memory feature to save this end point for further playback.

****Note:** Some Commodore software do not bother to disable cassette operation after the program has finished loading – this means a real cassette unit would continue playing the inserted cassette till the end of the tape was reached. Like-wise if the computer does not bother to de-assert the cassette motor control line that **SVI-CAS** relies on to switch between play and pause activities, then it will continue to play the rest of the image until the end of file is encountered. In this case the user must manually instruct **SVI-CAS** to stop playing via pausing or ending play-back.

11. ORIC-1 / ATMOS PLAYBACK

This section describes the general characteristics and operational specifics of **SVI-CAS** when used with the Oric-1 or Oric Atmos Computer.



Figure 11.1. Oric-1 / Atmos Operation

The Oric-1 and Oric Atmos computers like the majority of early 1980s computers made use of an external generic cassette unit for the loading of programs. To achieve best results it is recommended that the user makes use of a lead that not only connects audio to the computer, but also allows for the computer to control the playing and pausing of an image via tape motor control, some software requires this to load successfully.

If such a lead is not available it is still possible to connect only audio out from **SVI-CAS** to the computer, though the user will have to manually start the play-back of each image block by pressing  when prompted by **SVI-CAS** through its “**WAITING**” message.

1. Connect **SVI-CAS** to the computer cassette port, ensuring that the 3.5mm “EAR” jack of the lead is plugged into the **Tape OUT** port, the 3.5mm “MIC” jack of the lead is plugged into the **Tape IN** port, while also ensuring the 2.5mm jack is connected to the **Tape Motor Control** port. The DIN connector of the lead is plugged into the Oric computer cassette port. If unsure as to which sockets to use consult section 3.
2. Plug in the AC adapter and power-up **SVI-CAS**, ensuring it defaults to the main menu, then power-up the computer and ensure it boots correctly into BASIC.
3. Using **SVI-CAS** ensure that the correct interface is selected as “**Oric-1 & Atmos**” and that “**Automatic Play**” is enabled if the connecting lead supports it, else change to “**Manual Play**”.



Figure 11.2. Basic connections

4. Move to the directory menu and select the desired image to be loaded, be sure to select an image that is actually intended for this computer! Press  to move to the image inspection menu.
5. Select the desired entry inside the image which you would like to load, then press the  or  button. **SVI-CAS** will move to the play-back screen and display “**READY**” if it is waiting for the computer to initiate play-back, else it will display “**WAITING**” if the user is required to manually start play-back of the next image block.
6. On the computer enter the required command to start the loading process, enter the following command: **CLOAD”” <ENTER>**
7. After starting the loading process on the computer press the  button if loading manually when prompted to do so, else wait for the computer to finish loading the program. Watch the LCD display and check that the name of your selected entry appears after a short time, also verify that the computer also displays this same entry name as well during loading.

8. Throughout the loading process **SVI-CAS** will cycle between the “**PLAYING**” and “**PAUSE**” stages until the desired program has finished loading. If the loading process starts with a (Basic) file it may be necessary for the user to manually run this program in order for any further potential loading to proceed, as **SVI-CAS** will remain paused when the computer de-asserts the cassette motor signal.
9. After completion of loading a BASIC program the user should enter a “**RUN**” command to start the program if it does not start execution automatically.
10. When all required loading is completed and the program is up and running on the computer examine the state of **SVI-CAS**. If **SVI-CAS** is paused then this is usually because a program was loaded that was part of a compilation or there is extra data at the end of the image the computer may not require. Hence if **SVI-CAS** participation has ended, the user should exit back to the main menu.

****Note:** It needs to be stated that after examination of the 2012 TOSEC archive of Oric TAP images a number of images have been found to be faulty and hence will not load correctly when played as TAP images or as wave images and this includes attempting to use a media player for wave playback as well. The user will thus need to be aware of the potential for loading problems when using Oric TAP images from some archives or sources. In addition to this there are ROM differences between the Oric-1 and later Oric Atmos, the Oric-1 is noted to have several ROM bugs – some in relation to loading via the cassette interface.

12. ATARI PLAYBACK

This section describes the general characteristics and operational specifics of **SVI-CAS** when used with the Atari 400 / 800 / XL / XE series computers.



Figure 12.1. Atari 1200XL Operation

The Atari family of 8bit computers is unlike any other, making use of a specialized SIO port for all peripheral operations which included that of cassette operation. The cassette unit was also unique in that rather than performing generic playback and recording duties, it also utilized a conversion process to transform FSK signals to UART interpretable data.

Atari cassette based programs are stored a series of data chunks, each encapsulated via a basic framing protocol. When loading via standard ROM based cassette routines, the host computer will provide audible feedback in the form of pulsed tones for each chunk of data received. Should the loading process fail the frequency of feedback tones from the host computer will change to reflect this and alert the user to loading failure. It should also be noted that some commercial games use a custom loader which disables loading audio feedback with the Atari host computer remaining silent during the loading process – thus if a loading error occurs it may be much harder to detect until the entire image has been played.

SVI-CAS emulates Atari datasette operation via the use of a special Atari adapter cable required for connection to a host machine via the SIO port. The SIO port is keyed by the shape of the SIO plug, so cannot be fitted incorrectly, unlike Spectravideo or Commodore interface arrangements, the motor control of the Atari SIO port will not automatically enable if the play button on **SVI-CAS** is pressed at any time – the entry of a loading command on the host computer is specifically required before cassette motor control is enabled.

SVI-CAS is SIO compliant and will release the bus when not in use, so it is possible to daisy-chain **SVI-CAS** with other Atari peripherals without bus clashes occurring – though **SVI-CAS** must be idle and not “PAUSED” or “PLAYING” when attempting to use other devices like disk drives. Likewise, the user must wait for other peripherals to complete their operations and then release the bus before starting **SVI-CAS** operations.

SVI-CAS offers three baud rates for playback, 600 (Atari cassette standard), 800 , and 1000. It is recommended that the user first becomes familiar with **SVI-CAS** use at 600 baud – which is the standard Atari cassette speed, as higher baud rate selections may prove less agreeable with some Atari models. There are two general ways in which cassette images may be loaded into a host machine depending upon the image type, it must also be understood that unless the user knows before-hand what type of program is being loaded a trial & error approach will be required in respect of Atari commands as program images contain no information **SVI-CAS** can use to aid the user in this decision process.

The most common CAS images the user will encounter are either BASIC programs or self-booting binary files, the later requiring the host Atari computer to be instructed to auto-boot from the attached cassette unit (in this case **SVI-CAS**).

The following special adapter cable is required for Atari machines:



Figure 12.2. Atari Adapter Cable

1. Connect **SVI-CAS** to the computer SIO port ensuring that the correct orientation of the **SVI-CAS** SIO plug.

NEVER twist the cassette port plug to insert or remove it, this may result in damage or even serious electrical damage to both **SVI-CAS** and the host.

Connect the 3.5mm Stereo jack to the **Tape OUT** port, the 3.5mm Mono jack to the **Tape IN** port, 2.5mm jack to the **Tape Motor Control** port and insert the AC adapter power jack into the power socket. If unsure as to which sockets to use please consult section 3 before continuing.

NOTE: Attempting to connect **SVI-CAS** using an alternative means may result in damage to **SVI-CAS**, your computer, or both, this adapter cable is specially made for use with this image player and is more than simply a collection of leads & connectors.

If the user lacks (or has lost) this cable they should contact the author to arrange purchase via the same means with which **SVI-CAS** was originally obtained. Always check the plug key after removing the cable to ensure it is present and intact.



Figure 12.3. Atari Connection

2. Power-up **SVI-CAS**, ensuring it defaults to the main menu, ensuring that the correct interface and desired baud rate are selected as “**Atari <Baud>**” and that “**Automatic Play**” is enabled. In the case of Atari operation, it is strongly recommended that automatic playback control is used due to the nature of the Atari image format and SIO based cassette operations.
3. Move to the directory menu and select the desired image to be loaded, be sure to select an image that is actually intended for this computer, Press  to move to the image inspection menu.
4. To load BASIC programs use the “CLOAD” command and ensure that the BASIC programming language cartridge is installed if using an Atari 400 / 800 / 1200XL machine. Atari cassette based programs have no file name, therefore this command without any file name suffix is used for all BASIC program loading.

To auto-boot a binary CAS image the user must first disable BASIC and then force the host computer to “auto-boot” from the cassette unit, to do this perform the following based on the host machine type.

For the Atari 400 / 800 / 1200XL series machines

- a) While un-powered remove the BASIC programming language cartridge if present OR in addition to holding the START key down, also hold the OPTION key to disable BASIC when booting the machine.
- b) Hold the START key down and apply power to the host computer, the user should hear a short tone from the Atari. Release held key(s).
- c) Start **SVI-CAS** playing the desired CAS image and ensure that automatic tape motor control remains enabled using  or .
- d) When **SVI-CAS** displays “**PAUSED**” hit any key on the host computer to signal the Atari to start booting from the cassette image.
- e) If correctly done, after 20 seconds the user should start to hear short bursts of a medium tone which indicates the host machine is now loading the CAS image.

For the Atari XL & XE series machines (not 1200XL)

- a) Hold the START and OPTION keys down and apply power to the host computer, the user should hear a short tone from the Atari. Release held key(s).
 - b) Start **SVI-CAS** playing the desired CAS image and ensure that automatic tape motor control remains enabled using  or .
 - c) When **SVI-CAS** displays “**PAUSED**” hit any key on the host computer to signal the Atari to start booting from the cassette image.
 - d) If correctly done, after 20 seconds the user should start to hear short bursts of a medium tone which indicates the host machine is now loading the CAS image.
5. Throughout the loading process **SVI-CAS** will cycle between the “**PLAYING**” and “**PAUSE**” stages until the desired program has finished loading.
 6. After completion of loading a BASIC program the user should enter a “**RUN**” command to start the program if it does not start execution automatically. Auto-booted binary files will immediately start execution after they have finished loading.
 7. When all required loading is completed and the program is up and running on the computer examine the state of **SVI-CAS**. If **SVI-CAS** is paused then this is usually because a program was loaded that was part of a compilation or there is extra data at the end of the image the computer may not require. Hence if **SVI-CAS** participation has ended, the user should exit back to the main menu unless they plan to later load any further programs contained within the current image.
 8. If after only a few blocks played back, the Atari host machine displays an error, this is a good indication that the wrong loading choice was applied and the user should restart the process again using the other method, additionally if loading is still an issue, use of the default standard Atari baud rate of 600 baud should be considered as not all Atari models have been found to be equally reliable at higher baud rates.

13. GENERIC PLAYBACK

This section describes the general characteristics and operational specifics of **SVI-CAS** when used to playback an image in Generic Mode for additional computers not directly supported by **SVI-CAS** in one of its native modes. Generic playback allows the user to play wave, **TZX** and **SVI-CAS TAP** recordings for additional computers that are found to be compatible with this unit.



Figure 13.1. Generic Play Operation

Prior to using this mode with a non-directly supported computer it is suggested that the user familiarize themselves with the data loading specifics of the host machine in order to correctly adjust any of the main menu parameters that may be incompatible with the default settings **SVI-CAS** applies when this mode is first selected.

For the purpose of loading wave images **SVI-CAS** does not play them like a media player or program would, instead it employs a custom algorithm to generate a more “computer friendly” signal to help improve the chances of loading success. It must also be stated that the quality of the source wave image is also very important – low noise and an axis-symmetrical waveform will improve the chances of a successful loading operation.

When attempting to load a wave image, generic **TZX** image or actual **SVI-CAS TAP** recording, best results are obtained when the user makes use of a lead that not only connects audio to the computer, but also allows for the computer to control the playing and pausing of an image via tape motor control, software produced for some computers requires this level of control to load successfully. If the host computer does not support motor control, then free-play or manual loading will be the required main menu selection instead.

It should also be noted that in order to play a wave image on SVI-CAS, said image must meet the following specifications:

- **Frequency / sample rate** - *22050Hz, 16000Hz, 11025Hz, 9600Hz or 8000Hz*
- **Sample bit size** - *Unsigned 8 Bit PCM samples*
- **Track(s)** - *single / Mono track only*

Wave files that do not match these specifications will be rejected by **SVI-CAS**, though can be remastered with the correct parameters through tools such as Audacity.

For the purpose of loading generic **TZX** images or actual **SVI-CAS TAP** recordings the user must likewise also ensure appropriate main menu attributes are selected.



Figure 13.2. Basic connections

1. Connect **SVI-CAS** to the computer cassette port, ensuring that the 3.5mm “EAR” jack of the lead is plugged into the **Tape OUT** port, the 3.5mm “MIC” jack of the lead is plugged into the **Tape IN** port , while also ensuring the 2.5mm jack is connected to the **Tape Motor Control** port if the host machine has one. The DIN or individual leads are then plugged into the host computer cassette port ensuring the correct functionality is connected per lead. If unsure as to which sockets to use consult section 3.

2. Plug in the AC adapter and power-up **SVI-CAS**, ensuring it defaults to the main menu, then power-up the computer and ensure it boots correctly into BASIC.
3. Using **SVI-CAS** ensure that the correct interface is selected as “**Generic Mode**” and that “**Automatic Play**” is enabled if the connecting lead supports it, else enable free-play or manual-play via the disabling of all or some of the control methods.
4. Move to the directory menu and select the desired image to be loaded, be sure to select an image that is actually intended for this computer! Press  to move to the image inspection menu.
5. On the host computer enter the required command to start the loading process.
6. Select the desired entry inside the image which you would like to load, then press the  or  button. **SVI-CAS** will move to the play-back screen and display “**READY**” if it is waiting for the computer to initiate play-back, else it will display “**PLAYING**” if no playback control is enabled and image playback has commenced.
7. After starting the loading process watch the LCD display and check that the name of your selected entry appears after a short time, also verify that the host computer also acknowledges that loading is under way.
8. Throughout the loading process **SVI-CAS** will cycle between the “**PLAYING**” and “**PAUSE**” stages of the loading process should it be under computer control – indicating when assertion and de-assertion of the tape motor is taking place, else it will simply play the entire image through non-stop if automatic control is not available via the host machine.
9. When all required loading is completed and the program is up and running on the computer examine the state of **SVI-CAS**. If **SVI-CAS** is paused then this is usually because a program was loaded that was part of a compilation or there is extra data at the end of the image the computer may not require. **SVI-CAS** cannot determine where a complete program starts or ends.... It will continue to play further blocks of data till the overall end of an image file is reached if asked to do so. Hence if **SVI-CAS** participation has ended, the user should exit back to the main menu.

****Note:** The described loading process for wave images is based upon a very simple scenario, as it is impossible to predict the required handling of image loading for all computers and various software, it is up to the user to ensure they understand how to load a particular software package and execute the required command(s) in relation to the host machine.

14. HOW TO SPEED UP TSX IMAGE PLAY (MSX ONLY)

Unlike the **CAS** image format, the **TSX** format holds data describing the various timings required to reconstruct the original analog signal that would have been written to an old-school cassette tape. These timings define everything from the baud rate, to the length of any leader tone, and the duration of any pause (silence) after play-back of a data block completes.

Accordingly it has been generally stated within the retro community that that unlike **CAS** images, **TSX** images cannot be played at any desired baud rate – in reality under certain circumstances this is not exactly true specifically for MSX images, the key here being that the MSX standard allows for up to three (BIOS based) baud rates and the majority of MSX **TSX** images make use almost entirely of the 4B Kansas City data block in their composition.

SVI-CAS allows for the user to specify the baud rate as being automatically handled by the unit, or may be manually specified (overridden) by the user when loading **TSX** images. As **CAS** images are only manual in baud rate specification for MSX images this setting only applies to **TSX** image loading and is otherwise ignored. It should also be stated here that not all **TSX** images that use the **4B** block type can be loaded at a user set baud rate – two known game images, “Elite” & “Aliens” use this block, but employ custom loaders which require a specific baud rate for program loading and hence won’t work otherwise. Other game images that use the 4B block and employ BIOS based loaders can be user baud rate adjusted.

Images that consist of a mixture of different block types will be played thus:

- **4B** blocks played at user specified baud rate
- Non **4B** blocks will be played at their own specified baud rate

To set a higher baud rate for an MSX **TSX** image just select a higher MSX interface baud rate from the main menu and then select “Baud Manual”, then go through the steps to select the desired **TSX** image for play-back.

To save the user a lot of time and experimentation, a table is included in Appendix A, illustrating which known **TSX** images from various archives on the web can be loaded at higher baud rates and the level of benefit gained in terms of how much of an image comprises **4B** blocks. This table of results has been derived from using each image with **SVI-CAS** and tested across several MSX computers. It should also be noted that additional **TSX** imaged tapes are being added to these internet archives all the time, so the list may not be complete.

15. RECORDING WITH SVI-CAS

SVI-CAS is one of only a few devices currently available that is capable of recording from the host computer system directly to SD card. In doing so, **SVI-CAS** utilizes a range of emulator supported storage formats that are vastly superior to wave recordings in accuracy whilst occupying a fraction of the storage space, a typical file for a 64K recording will only be slightly larger than the actual program size in bytes as opposed to a wave recording of many megabytes and be technically superior in terms of accuracy as well.

In general, the following specific cassette image formats apply to recording with **SVI-CAS** for specifically supported computers:

Computer:	Recording Format:
Acorn	UEF (Chunk ID:100 & ID:110)
Amstrad	CDT (TZX Block ID:11)
Atari	CAS
Commodore	CBM TAP
Generic Mode	SVI-CAS TAP
Dragon	CAS
MSX	CAS
Oric-1 & Atmos	Oric TAP
Spectravideo (SVI)	CAS
Tandy CoCo	CAS
ZX80	“O” File
ZX81	“P” File
ZX Spectrum	TZX Block ID:10

Table 15.1. Recording Formats for Various Machines

This section describes the general process the user may follow to record their own programs from a host computer to SD card using **SVI-CAS**. The recording process is not limited to the specific machines directly supported, but extends to any machine that meets the signal specifications of **SVI-CAS**.

It should also be noted that in order to record a program image on SVI-CAS, the host computer must satisfy the following record signal specifications:

- **Recommended Max Baud Rate** - 2400 (single edge only, both edges for 1800 or less)
- **Record Signal Input Level** - 5 milli-Volts min to 2Volts max peak-peak

For the best possible and most size-efficient recordings it is strongly urged that the users makes use of any available tape motor control functionality if the host computer is equipped.

When selecting an interface from the main menu, **SVI-CAS** will select the correct edge trigger which is applied to the recording process, the only interface mode where the user may need to manually adjust the edge trigger is when making generic recordings. Under such a situation it is important for the user to set edge triggering correctly to avoid producing a non-working recording.

To record using **SVI-CAS** the following general steps apply:

On the host computer prepare the program to be saved and type in the relevant save command syntax for the machine, **but don't hit ENTER yet!**, general record command syntax for some machines include the following:

Commodore	SAVE “File Name”
MSX	CSAVE “File Name”, Baud Rate <i>Where Baud Rate = ‘1’ for 1200 and ‘2’ for 2400 baud</i>
Amstrad CPC	 TAPE [this selects tape mode for CPC6164 & 6128] SPEED WRITE Speed [optional] <i>Where speed = ‘0’ for 1000 and ‘1’ for 2000 baud</i> SAVE “File Name”
Dragon 32/64 ORIC-1 & ATMOS SVI 318/328 TRS80 CoCo TRS80 MC10 TRS80 Model 1 TRS80 Model 3 TRS80 Model 4	CSAVE “File Name”
ZX81 ZX Spectrum	SAVE “File Name” <i>Note: for the ZX81 the file name is not recorded but needed to satisfy syntax rules when saving</i>
Apple II/II+ ZX80	SAVE [file names are not supported]
Atari	CSAVE [file names are not supported]
Acorn	*TAPE [enables the cassette interface] SAVE “File Name”

Table 15.2. File Save Commands for Various Machines

It should be noted that several machines allow program saving at different baud rates, **SVI-CAS** supports baud rate detection for these machines and can generate recordings at their respective higher baud rates, though it is suggested for reliability the lower baud rate option (usually the machines default setting) is used.

In addition when using any interface setting other than Generic Mode, the standard tape protocol (ROM) based timings of the host computer must be used, **SVI-CAS** expects this in order to decode the incoming serial transmission from the host computer and produce a meaningful recording in the appropriate cassette image format, thus the user must avoid the use of higher speed custom recording routines unless Generic Mode is in use.

To initiate a recording, from the main menu screen press the “**RECORD**” button and navigate to the desired directory where the recording is to be saved. Navigation uses the same method as per selecting an image for playback, except that in record mode the user will not be allowed to open actual images, only sub-directories.



Figure 15.1. Record directory selection

Table 15.3. which follows, describes the button-based navigation of this particular menu level.

Function	Options
	Will return the user to the main menu
	- If a directory is highlighted this will drill down to the next directory level (maximum of eight nested levels) - If a file name is highlighted this function will be disabled (turns red).
	Accept the current directory for recording and move to the file name entry menu.
	- If within a sub-directory, this will drill up one directory level - If already at the root level of the directory, this will drop the user back to the main menu
	Move back one page of entries (if possible)
	Move back one displayed page entry (if possible)
	Move down one displayed page entry (if possible)
	Move forward one page of entries (if possible)

Table 15.3. Record directory Selection Buttons

When the desired directory has been reached press the  button to move to the file name entry process, here the user may enter the name of the file they would like to record, up to 250 characters in length. The name will scroll when it becomes too large to display the entire string on the display – it is recommended that the user applies discretion here and uses names of reasonable length rather than excessive, as they become harder to edit when they are several hundred characters long!

There is no need to add the file name suffix, **SVI-CAS** will do this automatically based of the recording image type.



Figure 15.2. File Name Entry Menu

During the file name entering process the user may prepare a file name via the displayed buttons as follows:

Function	Options
	Matrix of selectable file name characters.
	Return to the main menu.
	Accept entered file name and begin the recording process.
	Delete left-most character from file name.

Table 15.4. File Name Entry Process Buttons

To add another character to the file name press one of matrix character buttons  and to delete the left-most character use the  button. When the user is happy with their file name they should press the  button to move to the recording screen to commence the actual recording process.

When the recording screen is displayed **SVI-CAS** will begin to prepare for the recording process to begin, firstly the SD card will be checked and the available free space will be reported using a figure resolution in megabytes. The time taken to report free space can take longer for cards that are large and have badly fragmented file space, the user should avoid using a card in this condition as it will slow the writing speed and could impact upon recording response times.

The user should make sure there is at least one megabyte of free space available for recording to avoid the risk of the record process failing due to the SD card becoming full. During this SD checking process the message “**CHECKING CARD**” will be displayed, after which “**READY**” is displayed post-card checking.



Figure 15.3. Record Process Waiting for User

If tape motor control setting of the main menu is set to free-flow, i.e. both images are red, then **SVI-CAS** will automatically begin recording. Should the setting be manual, then **SVI-CAS** will reach a point where it is **“WAITING”** for the user to press the  button to begin the recording process, otherwise, if it is automatic, then **SVI-CAS** will **“PAUSE”** and wait for the host computer to assert the tape motor control line which will then allow **“RECORDING”** to start. The user should hit enter on the host computer when either **“PAUSE”** or **“RECORDING”** are first displayed to begin the program recording.



Figure 15.4. Record Process Underway

During the recording process the user may pause the recording manually and restart it via the displayed enabled buttons as follows:

Function	Options
	Pause record process, disabled if button text is RED in colour.
	Begin or continue recording process, disabled if button text is RED in colour.
	End and close out the current recording, disabled if button text is RED in colour.

Table 15.5. Active Recording Process Buttons

If the process is under automatic control, the host computer may elect to stop and restart recording as it sees fit. When the host computer has finished recording the user may either pause the recording process if they want to save another program straight after the current one in the same image and then restart recording..... or they may press the stop button which results in the recorded image being closed out.



Figure 15.5. Record Process Successfully Completed

Post recording **SVI-CAS** displays a new set of buttons that allow the user to return to the main menu, open the current recording directory where the newly recorded image has been saved, or go back to the file name menu where the user may enter a new file name and then make another recording.

Function	Options
	Return to the main menu.
	Open the current directory where the recorded image was saved.
	Go back to the file name entry menu.

Table 15.6. Completed Recording Process Buttons

In summary, the recording process in its simplest form may be expressed as the flow chart shown in **Figure 15.6.** that follows.

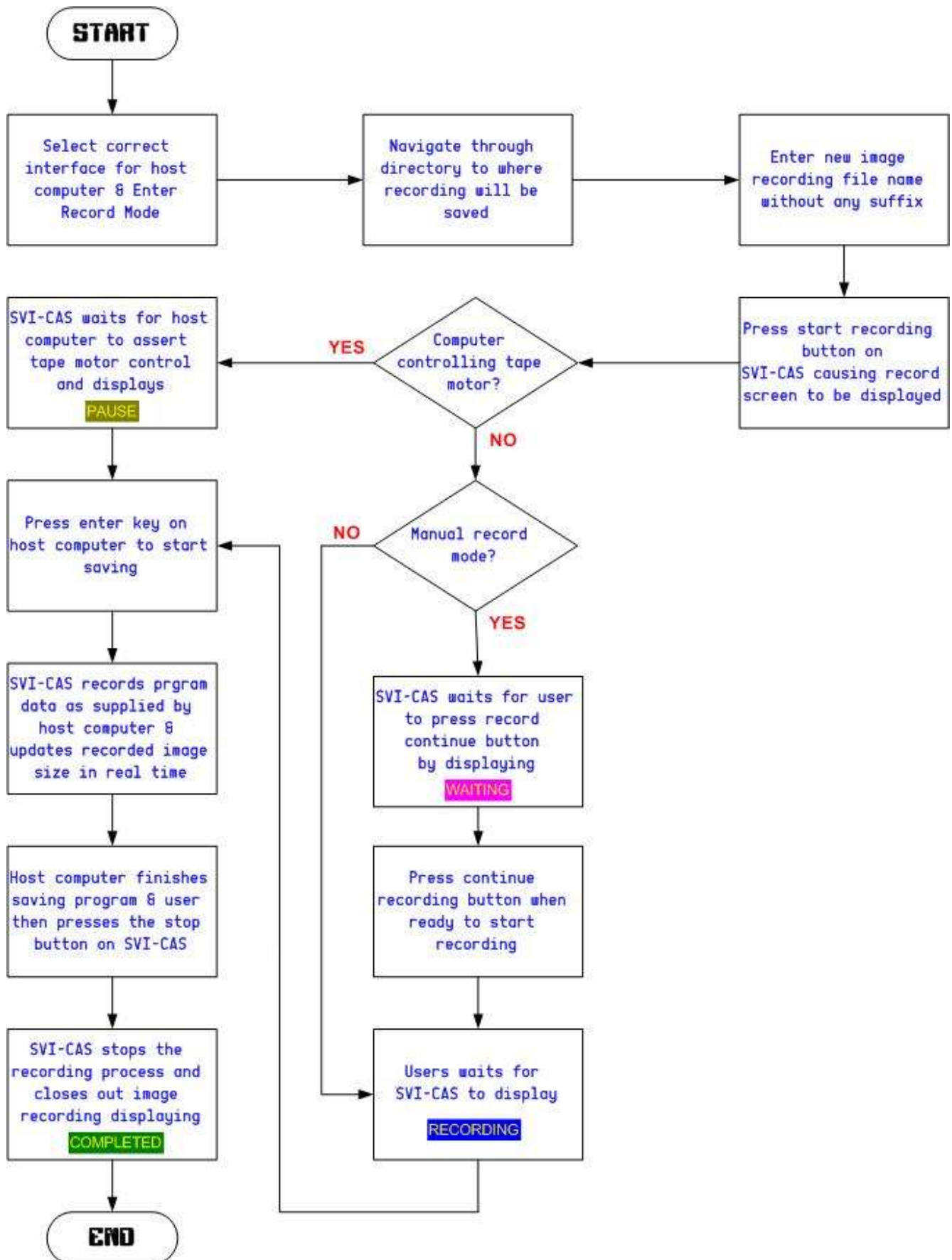


Figure 15.6. Recording Process Simplified

15.1 Recording Errors

Like any recording process there is always the risk of an error occurring that brings the process to a failed end. **SVI-CAS** has a collection of status messages that are used to give the user a brief indication of what stage the recording process is currently at and to alert the user to a possible reason why an error during the recording process has occurred. A brief list and explanation of each of the possible messages follows:

Status Message	Explanation
File Open	A new file has been created and header data written.
File Closed	The current image recording has been successfully closed out.
Path Error	The current path is invalid, there is an issue with the directory or FAT, or the SD card has been hot-swapped with another which does not have the same directory structure as the previous one after entering the file name for recording.
File Exists	A recording was attempted using the name of an existing file, enter a new file name or delete the existing file.
Open Error	A new file could not be opened for writing on the installed SD card, check the card for incorrect installation in the SD card slot, card may also be faulty / worn.
Seek Error	The desired point with a currently open file could not be reached, this may be due to a number of factors including issues with SD card installation, card reliability, card may also be faulty / worn.
Writing Data	The currently open file is being successfully written to by SVI-CAS .
Write Paused	Writing to the currently open file is currently paused by SVI-CAS .
Write Error	The currently open file has developed an unknown error during writing, check the SD card for similar issues as previously described for other faults.
Other Error	An internal error has occurred that has caused a failure in the recording process that can't be specifically identified.

Table 15.1.1. Recording Status Messages

One specific issue that can impact upon recording is that of the host computer's record signal which normally connects to the microphone socket of an analogue tape recorder, as previously stated the minimum signal amplitude must be 5 millivolts peak-peak in order for **SVI-CAS** to successfully amplify and process it. It should also be noted that **SVI-CAS** cannot tell the difference between random pulses of noise and actual data, most of this noise is never recorded when the host computer is able to control the recording process via the tape motor control line, but computers such as the ZX Spectrum have no such ability, so this noise may be audible when listening to a recording – but has no effect on the quality of the recording or playback.

After completing a recording it is suggested the user enable the internal speaker during playback and then listen to this new recording to confirm that it sounds as they would expect for a given computer, most classic home computers have specific pilot tone and data / bit frequencies that are readily identifiable to experienced computer users.

15.2 Possible Recording Approaches

The approach the user takes towards a recording can be one of three basic methods which all depend upon the main menu tape motor control setting. **These approaches are for non-Commodore or Spectravideo 318/328 computers.**

Tape Motor Control = Disabled (Free Running Mode)

In this instance when the user presses the  button in the file name entry menu, recording will start immediately without any pause or delay, thus the user should first prepare the host computer and type in the save command required, **but don't hit ENTER yet!**

The user simply waits for the “**RECORDING**” message to be displayed and then hits enter on the host computer to start the recording process. The size of the growing image in bytes will be displayed below this message and update regularly as recording proceeds.

When the host machines informs the user that program saving is completed the user should hit either the  button to stop and complete the recording, or the  button if they would like pause the recording to add another program to the currently open image.

Note that in this mode any noise generated by the host computer will cause the addition of further bytes to the image after the actual program has finished saving, so it is important to end the recording process ASAP after the host computer has completed saving. This method is most useful for the Sinclair range of supported machines.

Tape Motor Control = Manual (User Initiated Mode)

In this instance when the user presses the  button in the file name entry menu, recording will open a new image for writing but stop and wait for the user to press the  button before the recording of actual data begins, thus the user should first prepare the host computer and type in the save command required, **but don't hit ENTER yet!**

The user now presses the  button and waits for the “**RECORDING**” message to be displayed, as soon as this message is displayed hit enter on the host computer to start the recording process. The size of the growing image in bytes will be displayed below this message and update regularly as recording proceeds.

When the host machines informs the user that program saving is completed the user should hit either the  button to stop and complete the recording, or the  button if they would like pause the recording to add another program to the currently open image. The difference between this mode and the previous mode concerns the ability for the user to control when **SVI-CAS** starts recording without the need for tape motor control support.

Tape Motor Control = Automatic (Host Computer Control Mode)

In this instance when the user presses the  button in the file name entry menu, recording will open a new image for writing but pause and wait for the host computer to assert the tape motor control line before the recording of actual data begins, thus the user should first prepare the host computer and type in the save command required, **but don't hit ENTER yet!**

The user now prepares **SVI-CAS** as per usual, entering a file name and then pressing  to start the saving of a program. **SVI-CAS** will now proceed to a point where it will “**PAUSE**” and wait for the host computer to start the saving process, now the user hits enter on the host computer to start saving and when the tape motor control is asserted **SVI-CAS** will start recording. During this process **SVI-CAS** may alternate between the “**RECORDING**” and “**PAUSE**” messages. The size of the growing image in bytes will be displayed below this message and update regularly as recording proceeds.

When the host computer has completed program saving, it will leave **SVI-CAS** in a paused state. Should the user want to add another program to this image they should press the  button to manually pause the record process and then prepare the host machine with another program to save and then repeat the whole process, otherwise the user should press the  button to end the recording process and close out the recorded image. Recordings made using automatic control of the whole process are the most space efficient and noise free of the three available options.

Note that this is the preferred method of the three if the host computer supports it, Commodore & SVI-318/328 machines are an exception where the user may actually press enter on the host computer first as the host machine will wait for SVI-CAS to assert the play / record buttons at the start of recording before actually starting to send SVI-CAS data for recording purposes. This is because these two machines have the capability to monitor the logical state of the cassette units they use – a feature that other supported computers lack due to their use of generic analog cassette recorders.

15.3 Recording with the ZX80 & ZX81

The Sinclair ZX80 and ZX81 computers are unique in the retro-computing world for a number of reasons, one being the use of their composite video sync signal for the purpose of recording programs to cassette. Unlike other computers of the era these two Sinclair machines when saving a program will actively stop generating a video display and instead switch to the generation of pulses that represent program data, this is why the user sees the display on their monitor / TV become a mess of B&W horizontal bars during saving and loading on the ZX81. The ZX80 also generates such a display when saving, but the display remains black during loading.

To describe this visually:

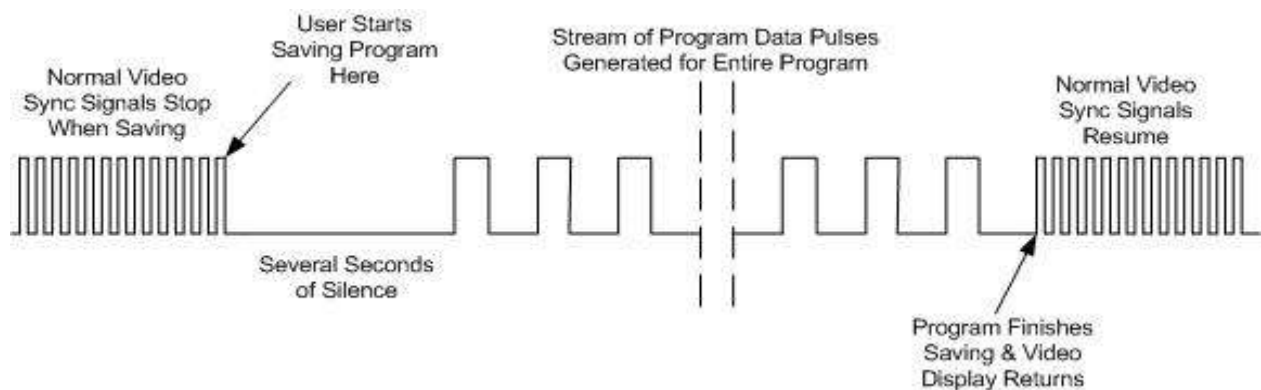


Figure 15.3.1. Typical ZX80 / ZX81 Recording Signal

This is why the user will hear a buzzing sound recorded from the MIC output of their ZX80 or ZX81 before the actual saving process starts (video display is still being generated) and also hear this buzzing sound straight after the save process has completed (signifying a return to video display generation again).

Accordingly since these machines lack tape motor control capabilities it is suggested to run the record operation in free-mode so it starts without any delays, pre-type in the save command on the computer and wait for **SVI-CAS** to display **“RECORDING”** as soon as this displays, press enter on the ZX80/81 to start saving and as soon as the video display of the ZX80/81 returns, press ☐ or ☐ buttons on **SVI-CAS** to end the recording.

An important thing to also note with these two computers concerns modifications by users, some may choose to bypass the RF modulator and directly connect the video signal that is normally fed to the modulator to a composite monitor. This signal should be buffered with a small video amplifier stage otherwise the signal will suffer a loss of amplitude as the current output of the ULA of the ZX81 for video generation is weak – this loss of signal amplitude will make the signal too weak for **SVI-CAS** or even an analogue tape recorder to effectively record.

The ZX80 is only slightly better since it comprises discrete TTL, but the issue still remains and proper buffering is required should the user choose this path for display purposes.

The ZX Spectrum unlike its two predecessors, does not suffer from any of the above issues in spite of also having no tape motor control facilities. Another issue that can affect Sinclair machines (the ZX80 & ZX81 in particular) is that of electrical noise from the television or CRT monitor being used or noise from the internal oscillator of the Sinclair 16k ram pack.

Thus to ensure high quality recordings it is best practice to keep the tape EAR & MIC leads as far away from the ram pack and television as is reasonable. Also avoid leads that are excessively long, use the original tape lead supplied with the machine if the user possess it. Users have found it beneficial when experiencing recording problems to unplug the ear connection from these machines to reduce the effects of earth loops and other potential noise.

The ZX80 & ZX81 are very susceptible to external noise due to their cheap design and almost non-existent shielding. Failure to observe these precautions may produce un-loadable recordings for these machines. It is suggested that the user experiments with recordings and determines what works best for them before attempting to save actual program data.

If ZX80/81 recording in Generic Mode instead, when playing back a recording the user has made it can be beneficial to enable the internal **SVI-CAS** speaker to listen as the recording is played. Usually there will be a short period of buzzing heard (video sync still being generated) which needs to be played-past to reach the several seconds of silence before actual program data beings. This buzzing sound can upset the ZX80 during loading so it is best to play past it to the silence and then start loading on the ZX80 when that silence has been reached, the ZX81 has been found to be somewhat more tolerant of this pre-silence noise and tends to load on most occasions regardless of the user playing-past it or not.

15.4 General Recording Points

The recording process used by **SVI-CAS** relies on the timed monitoring of pulse transitions through an edge detection process when determining bit polarities during image recording, the monitoring of these edges may be positive, negative or both. The recording system used by **SVI-CAS** allows for very accurate recordings to be made that consume little storage space, yet produce quality playback results.

Generic Mode images that are created where both edges are monitored are double the size of an image that is only created using one edge or the other – some vintage computers operate on a positive pulse cycle when starting their recordings, others on a negative cycle, there is though, a third case where a small number of machines randomly start with one cycle or the other and may even change phase 180 degrees after each byte has been transmitted. Under such conditions a recording is just be made using both edges, such machines include the Apple II / Ile

Due to the limitations of the SVI-CAS 8-bit CPU hardware and maximum SPI burst write speed (4MHz), baud rates in excess of 1800 baud should not be recorded with both trigger edges enabled, as this will generate more data than can be written to the SD card and result in non-working recordings with large patches of silence due to SD card write operations not able to keep up with the high data rate. 99.9% of computers (includes all famous well known machines) only require single edge recording, while the other 0.1% operate at low baud rates, so this is not considered a disadvantage the user may encounter.

For machines that have specific selectable interface entries with **SVI-CAS** the user does not need to even concern themselves with this issue, as **SVI-CAS** will automatically select and configure the recording requirements without the need for user intervention, the user will need to be careful not to change these default settings if they are going to make a recording.

If they do and can't remember what the original setting was, just cycle through the interface options again to the same host computer to restore the settings back to their defaults. For the new user, it is suggested they start out with a blank SD card and experiment saving small programs from their computers to come to grips with the recording process, it is not difficult to master in a short period of time. It is also recommended that the user keep their own programs separate from any directories of other images they may have.

Never attempt to remove an SD card being written to by **SVI-CAS**, as this may cause damage to the file allocation table of the card and cause file loss or even make the card completely unreadable and only recoverable in terms of reformatting.

Depending upon the mode of recording (host computer controlled or manual / free-running), recording may continue on after the host machine has finished saving because noise is detected by **SVI-CAS** from the host computer, hence it is desirable to end a recording asap after the host machine has finished if the host machine is not able to pause the recording operation by its self. This is not an issue if the host machine is noisy, yet has tape motor control functionality, as it will pause the recording anyway.

Lastly, while every effort has been made to provide a reliable means with which to record programs to SD card, the record process involves an analog signal being amplified, filtered, shaped and then written to a storage medium – during this multistage process errors are possible, be they electrical noise from a number of potential external sources, some form of disturbance to **SVI-CAS** while recording, or even some form of SD card failure. Hence the user must accept that there is no method available that can be said to be 100% reliable. Careful execution of the recording process by the user as well as looking after of equipment will help minimize the chance of potential failure.

15.5 Atari Recording Specifics

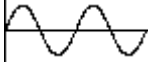
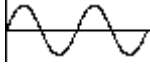
After starting the CSAVE operation on an Atari computer it will play two warning tones to indicate it is now ready to begin saving, at this point when **SVI-CAS** is ready to start recording the user should then strike any key on the Atari's main keyboard and the recording process will then commence.

The Atari recording process will:

- a) Produce two tones to indicate it is ready to record.
- b) Require the user to then press any key on the main keyboard to start the process.
- c) Generate 20 seconds of silence at the start of the recording.
- d) Continue to provide audible feedback while recording via a pulsed tone for every block the Atari generates.
- e) At the end of the recording the Atari will remain silent and **SVI-CAS** will indicate "**PAUSE**".

15.6 Using Other Computers with SVI-CAS

The playback and recording processes used by **SVI-CAS** are open to use by any machine that possesses a cassette interface and also meets the signal requirements for recording as stated early in this chapter. To determine the specific settings required by a particular machine it is suggested that the user use the following general plan and in doing so be able to save their own programs and load them back in using this product.

1. Start with a playback phase of 
2. Start with a recording edge trigger of 
3. Type a few lines of code into the host computer and then attempt to save this code as a recording on **SVI-CAS**
4. Now attempt to playback this recording and load it back into the host computer, if it works then make a note of these settings and ensure they are set when using “Generic Mode” with this machine, else invert both the playback phase and the record edge triggering and try again.
5. If this new recording also won't load then select a playback phase of  again and this time set the record edge triggering to  and try again to save a small program and then reload it back in.
6. If again unsuccessful it is finally suggested that cables as well as the output record signal of the host computer are checked to ensure they are compatible with **SVI-CAS**.

To aid the user, **Appendix C** presents a table containing **Generic Mode** settings that **SVI-CAS** users have been able to verify as being suitable for various other machines not directly supported by **SVI-CAS**.

APPENDIX A. TSX IMAGE LOADING RESULTS

The following table details the use and actual achieved results of **SVI-CAS** high-speed (3600 baud rate) loading of **TSX** images into three specific MSX computers, a Yamaha CX5M (32k with 64K expander fitted), a SVI-728 (64K) and an SVI-738 Xpress (64K) machine. Where an image has been found to not load successfully into any of these machines it has been converted to a wave file for a final loading check, if still failing – is marked accordingly. Percentage benefit is a rough guide as to how much of the image will load at a manually specified higher baud rate.

TSX Game Image	% Benefit	CX5M	SVI-728	SVI-738
007 A View to a Kill	100	✓	✓	✓
007 License to Kill	10	✓	✓	✓
007 The Living Daylights	100	✓	✓	✓
10th Frame	100	✓	✓	✓
180 (Darts)	100	✓	✓	✓
3D Golf Simulation	100	✓	✓	✓
4x4 Off-Road Racing (side A)	1	✓	✓	✓
4x4 Off-Road Racing (side B)	0	✓	✓	✓
6 in one	100	✓	✓	✓
737 Flight Simulator	100	✓	✓	✓
747 Flight Simulator	100	✓	✓	✓
A.R.C.O.S.	100	✓	✓	✓
Aaargh!	10	✓	✓	✓
Aacko Draw & Paint	100	✓	✓	✓
Abadia del Crimen	100	✗	✓	✓
Abu Simbel Profanation (Clam case)	100	✓	✓	✓
Abu Simbel Profanation (Jewel case)	100	✓	✓	✓
Ace of Aces	100	✓	✓	✓
Adres MSX	100	✓	✓	✓
After The War (side A)	100	✗	✓	✓
After The War (side B)	100	✗	✓	✓
Afteroids	100	✓	✓	✓
Alcatraz - A fuga impossivel	100	✓	✓	✓
Ale Hop!	100	✓	✓	✓
Alien 8	100	✓	✓	✓
Alien Syndrome (side A)	100	✗	✓	✓
Alien Syndrome (side B)	100	✗	✓	✓
Aliens – Must be played at encoded baud rate !!!	0	✓	✓	✓
Angel Nieto Pole 500cc	100	✗	✓	✓
The Apeman, Strikes Again	100	✓	✓	✓
Aprende Astronomia, El Firmamento	100	✓	✓	✓
Aprende Astronomia, La Luna	100	✓	✓	✓
Aprende Astronomia, La Tierra	100	✓	✓	✓
Aprende Astronomia, Los Planetas	100	✓	✓	✓
Arkanoid – disable disk ROM at boot up	100	✗	✓	✓
Army Moves (side A)	100	✓	✓	✓
Army Moves (side B)	100	✓	✓	✓
Aquimedes XXI – (Basic loader has syntax error)	100	✗	✓	✓
Aspar GP Master	100	✓	✓	✓
Auf Wiedersehen Monty	1	✓	✓	✓

Autocrash	100	✓	✓	✓
Avenger	1	✓	✓	✓
Aventura Espacial, La (side A)	100	✓	✓	✓
Aventura Espacial, La (side B)	100	✓	✓	✓
Aventura Original, La (side A)	100	✓	✓	✓
Aventura Original, La (side B)	100	✓	✓	✓
Averno	100	✓	✓	✓
B.C. Bill II - Grog's Revenge	100	✗	✓	✓
BMX Simulator – disable disk ROM at boot up	100	✗	✓	✓
Bang! Bang!	1	✓	✓	✓
Barbarian	1	✓	✓	✓
Barbarian II	1	✓	✓	✓
Barnstormer	100	✓	✓	✓
Base de Datos (Database)	100	✓	✓	✓
Batalla de Tanques	100	✓	✓	✓
Batman	100	✓	✓	✓
Batman the Movie (Side A)	1	✓	✓	✓
Batman the Movie (Side B)	0	✓	✓	✓
Battle Chopper	100	✓	✓	✓
Beach-Head – disable disk ROM at boot up	100	✗	✓	✓
Beamrider	100	✓	✓	✓
Black Beard (English)	10	✓	✓	✓
Black Beard (Spanish)	10	✓	✓	✓
Black Jack	100	✓	✓	✓
Blagger (Spanish)	100	✓	✓	✓
Blagger (English)	100	✓	✓	✓
Blasteroids	1	✓	✓	✓
Blow Up!	100	✓	✓	✓
The Blue Wolf and the White Stag (Side A)	100	✓	✓	✓
The Blue Wolf and the White Stag (Side B)	100	✓	✓	✓
Boardello	100	✓	✓	✓
Booga-Boo the Flea	100	✓	✓	✓
Boom!	100	✓	✓	✓
Boulder Dash (English)	100	✓	✓	✓
Boulder Dash (Spanish)	100	✓	✓	✓
Bounce	100	✓	✓	✓
Bounder (Clam Case)	100	✓	✓	✓
Bounder (Jewel Case)	100	✓	✓	✓
Bousou Tokkyuu SOS, Stop the Express	100	✓	✓	✓
Break Out! – will not work at all	0	✗	✗	✗
Bronx	1	✓	✓	✓
Bruce Lee	100	✗	✓	✓
Bubbler	100	✓	✓	✓
Buck Rogers	100	✓	✓	✓
Camelot Warriors (Clam Case)	100	✓	✓	✓
Camelot Warriors (Jewel Case)	100	✓	✓	✓
Cannon Fighter	100	✓	✓	✓
Castle Combat (Clam Case)	100	✓	✓	✓
Castle Combat (Jewel Case)	100	✓	✓	✓
Checkmate! – will not work at all	0	✗	✗	✗
Chess'86	100	✓	✓	✓

Chicago 30's	10	✓	✓	✓
Chiller	100	✓	✓	✓
Chima Chima	100	✓	✓	✓
Ci-U-Than Trilogy II - Los Templos Sagrados (Side A)	1	✓	✓	✓
Ci-U-Than Trilogy II - Los Templos Sagrados (Side B - Buggy)	1	✓	✓	✓
Ci-U-Than Trilogy II - Los Templos Sagrados (Side B - Fixed)	1	✓	✓	✓
El Cid – disable disk ROM at boot up	100	✗	✓	✓
Classic Adventure	100	✗	✓	✓
Cobra's Arc	100	✓	✓	✓
Cocina Moderna	100	✓	✓	✓
Coliseum	100	✗	✓	✓
Colony	100	✓	✓	✓
Colt 36	100	✓	✓	✓
Comando Quatro	100	✓	✓	✓
Cometa Halley	100	✓	✓	✓
Compray Vende	100	✓	✓	✓
Computer Scrabble	100	✓	✓	✓
Contabilidad Domestica	100	✓	✓	✓
Corsarios (Side A)	100	✗	✓	✓
Corsarios (Side B)	100	✗	✓	✓
Cosmic Sheriff	100	✓	✓	✓
Crazy Golf	100	✓	✓	✓
Cubit	100	✓	✓	✓
Curro Jimenez	100	✗	✓	✓
Cyberbig	1	✓	✓	✓
Cyberun (Spanish)	100	✓	✓	✓
Cyberun (English)	100	✓	✓	✓
The Dam Busters	100	✗	✓	✓
Darts	100	✓	✓	✓
Dawn Patrol (spanish)	100	✓	✓	✓
Dawn Patrol (English)	100	✓	✓	✓
De Nachtwacht	100	✓	✓	✓
Death Wish 3 (spanish)	1	✓	✓	✓
Death Wish 3 (English)	1	✓	✓	✓
Decathlon	100	✓	✓	✓
Defcom 1	1	✓	✓	✓
Demonia	100	✓	✓	✓
Desolator	100	✓	✓	✓
Desperado	1	✓	✓	✓
Desperado 2	1	✓	✓	✓
The Devil's Castle	100	✓	✓	✓
Disc Warrior	100	✓	✓	✓
Discovery	100	✓	✓	✓
Dizzy Balloon	100	✓	✓	✓
Donkey Kong	100	✓	✓	✓
Double Dragon II - The Revenge	1	✓	✓	✓
Drakkar	10	✗	✓	✓
Drazen Petrovic Basket	100	✓	✓	✓

Driller Tanks	100	✓	✓	✓
Drome	100	✓	✓	✓
Dustin	100	✓	✓	✓
Eddie Kidd Jump Challenge	100	✓	✓	✓
Eggy	100	✓	✓	✓
Elite – Must be played at encoded baud rate !!!	0	✓	✓	✓
Emilio Butragueno Futbol	1	✓	✓	✓
Emilio Butragueno Futbol II (Siade A)	1	✓	✓	✓
Emilio Butragueno Futbol II (Side B)	1	✓	✓	✓
Enchanted	100	✗	✓	✓
Fernando Martin Basket Master	100	✓	✓	✓
Fernando Martin Basket Master Executive	100	✓	✓	✓
Fernando Martin Basket Master Executive (Alternative release)	100	✓	✓	✓
Feud	100	✓	✓	✓
Final Countdown	100	✓	✓	✓
Finders Keepers	100	✓	✓	✓
Fire Star	100	✗	✓	✓
Fire Hawk	100	✓	✓	✓
Flash Gordon	100	✓	✓	✓
Les Flic	100	✓	✓	✓
The Flintstones	1	✓	✓	✓
Football Manager - World Cup Edition	100	✓	✓	✓
Foton 4	100	✓	✓	✓
Frankie	100	✓	✓	✓
Freddy Hardest (Side A) (English)	100	✓	✓	✓
Freddy Hardest (Side B) (English)	100	✓	✓	✓
Freddy Hardest (Side A) (Spanish)	100	✓	✓	✓
Freddy Hardest (Side B) (Spanish)	100	✓	✓	✓
Freddy Hardest in South Manhattan	100	✓	✓	✓
Future Knight	1	✓	✓	✓
Game ABC - Introduction to Game Programming	100	✓	✓	✓
Gauntlet (Side A)	100	✓	✓	✓
Gauntlet (Side B)	100	✓	✓	✓
El Gerente	100	✓	✓	✓
Ghostbusters (Japan)	100	✓	✓	✓
Ghostbusters (US)	100	✓	✓	✓
Ghostbusters II (Spanish)	1	✓	✓	✓
Ghostbusters II (Alternate version) (Spanish)	1	✓	✓	✓
Goody	100	✗	✓	✓
Gunfright	100	✓	✓	✓
Guttblaster	100	✓	✓	✓
H.E.R.O.	100	✓	✓	✓
Haunted House	100	✓	✓	✓
Head over Heels	100	✓	✓	✓
Hercules, Slayer of the Dawned	1	✓	✓	✓
HiSoft Pascal MSX	100	✓	✓	✓
Hopper	100	✓	✓	✓
Howard The Duck – will not work at all	0	✗	✗	✗
Humphrey	100	✓	✓	✓
Hunchback	100	✓	✓	✓

Hundra	100	✓	✓	✓
Hype	100	✓	✓	✓
Ice	100	✓	✓	✓
Inca	100	✓	✓	✓
Indiana Jones and the Temple of Doom (Spanish)	100	✓	✓	✓
Indiana Jones and the Temple of Doom (English)	100	✓	✓	✓
Los Inhumanos	100	✗	✓	✓
International Karate	100	✓	✓	✓
Jabato (Side A)	100	✓	✓	✓
Jabato (Side B)	100	✓	✓	✓
Jack the Nipper (Clam case)	1	✓	✓	✓
Jack the Nipper (Jewel case)	1	✓	✓	✓
Jeca Bomba	100	✓	✓	✓
Jet Set Willy	100	✓	✓	✓
Jet Set Willy II	100	✓	✓	✓
Johnny Comomolo in 3,2,1 Fire – disable disk ROM at boot up	100	✗	✓	✓
Jump	100	✓	✓	✓
Ke Rulen los Petas	1	✓	✓	✓
Khazzad-Dum	100	✓	✓	✓
Klax	1	✓	✓	✓
Knight Lore	100	✓	✓	✓
Knight Tyme – disable disk ROM at boot up	100	✗	✓	✓
Krakout	1	✓	✓	✓
Krypton	100	✓	✓	✓
The Last Mission	100	✓	✓	✓
Lazer Bykes	100	✓	✓	✓
Lazy Jones	100	✓	✓	✓
Le Mans	100	✓	✓	✓
Livingstone Supongo	100	✓	✓	✓
Livingstone Supongo (Alternate version)	100	✓	✓	✓
Mad Mix 2	100	✓	✓	✓
Mad Mix	100	✓	✓	✓
Mad-Fox	100	✓	✓	✓
Manic Miner	100	✓	✓	✓
Map Game	100	✓	✓	✓
Mask II	100	✗	✓	✓
Mask III - Venom Strikes Back	1	✓	✓	✓
Masters Of The Universe	1	✓	✓	✓
Maziacs	100	✓	✓	✓
Metropolis	1	✓	✓	✓
El Misterio del Nilo	100	✓	✓	✓
Mortadelo y Filemon	1	✗	✓	✓
Mortadelo y Filemon 2 Safari Callejero (Side A)	1	✓	✓	✓
Mortadelo y Filemon 2 Safari Callejero (Side B)	1	✓	✓	✓
MSX Games Vol 04 - Xider	100	✓	✓	✓
El Mundo Perdido	1	✓	✓	✓
Mythos	100	✗	✓	✓
Night Knight – disable disk ROM at boot up	100	✗	✓	✓
Nightshade	100	✗	✓	✓
Nightshade (Alternate version)	100	✗	✓	✓

Ninja Kage	100	✓	✓	✓
Nonamed	100	✓	✓	✓
North Sea Helicopter	100	✓	✓	✓
Nuclear Bowls	100	✓	✓	✓
Obliterator	1	✗	✓	✓
Ocean Conqueror	100	✓	✓	✓
Octagon Squad	100	✓	✓	✓
Oh Shit!	100	✓	✓	✓
Operation Wolf (Side A)	100	✓	✓	✓
Operation Wolf (Side B)	100	✓	✓	✓
Ormuz	100	✗	✓	✓
Out Run	1	✓	✓	✓
Panzer Attack	100	✓	✓	✓
Pastfinder	100	✓	✓	✓
Penguin	100	✓	✓	✓
Perico Delgado Maillot Amarillo	1	✗	✓	✓
Phantis (Side A)	100	✓	✓	✓
Phantis (Side B)	100	✓	✓	✓
Phantomas II	100	✗	✓	✓
Piso Zero	100	✓	✓	✓
Pitfall II - Lost Caverns	100	✓	✓	✓
El Poder Oscuro	100	✗	✓	✓
Polar Star	100	✗	✓	✓
Post Mortem	100	✓	✓	✓
Puddles (Tape 1)	100	✓	✓	✓
Puddles (Tape 2)	100	✓	✓	✓
Puddles (Tape 3)	100	✓	✓	✓
Punch and Judy – will not work at all	0	✗	✗	✗
R.A.M.	1	✓	✓	✓
River Raid	100	✓	✓	✓
Robocop (Side A)	100	✓	✓	✓
Robocop (Side B)	100	✓	✓	✓
Rock 'n Roller	100	✓	✓	✓
Rocky	100	✓	✓	✓
Sabotaje	100	✓	✓	✓
Sabrina	100	✓	✓	✓
Samantha Fox Strip Poker	100	✓	✓	✓
Science Fiction	100	✓	✓	✓
Score 3020	1	✓	✓	✓
Scramble Spirits (Side A)	1	✓	✓	✓
Scramble Spirits (Side B)	1	✓	✓	✓
Sea Hunter (Version 1)	100	✓	✓	✓
Sea Hunter (Version 2)	100	✓	✓	✓
Sea Hunter (Version 3)	100	✓	✓	✓
Sea King	100	✓	✓	✓
El Secreto de la Piramide	100	✓	✓	✓
Silent Shadow	1	✓	✓	✓
Silfi	100	✓	✓	✓
Sky Vision	100	✓	✓	✓
Skyhawk	100	✓	✓	✓
Snake	100	✓	✓	✓

Sorcery	1	✓	✓	✓
Soul of a Robot	100	✓	✓	✓
Space Busters	100	✓	✓	✓
Spirits	100	✓	✓	✓
Spitfire 40	100	✓	✓	✓
Star Fighter	100	✓	✓	✓
Star Runner	100	✓	✓	✓
Star Wars	100	✓	✓	✓
Stardust	1	✓	✓	✓
Starquake	100	✓	✓	✓
Stars MSX - Issue 04 Extra Navidad	100	✓	✓	✓
Storm - Una's Lair	100	✓	✓	✓
Stormbringer – will not work at all	0	✗	✗	✗
Super Chess	100	✓	✓	✓
Survivor	100	✓	✓	✓
T.N.T.	100	✓	✓	✓
Tai-Pan (Side A)	100	✓	✓	✓
Tai-Pan (Side B)	100	✓	✓	✓
Target Plus	100	✓	✓	✓
Tasword	100	✓	✓	✓
Tawara	100	✓	✓	✓
Teach Yourself Basic	100	✓	✓	✓
Temptations	1	✓	✓	✓
Terminus – will not work at all	0	✗	✗	✗
Terrorpods	1	✓	✓	✓
Tetris (Spanish)	100	✓	✓	✓
Tetris (English)	100	✓	✓	✓
Thexder	100	✓	✓	✓
Thing Bounces Back	100	✓	✓	✓
Thor	100	✓	✓	✓
Thunder Blade	1	✓	✓	✓
Tiburón, aka JAWS	1	✓	✓	✓
Toobin	1	✓	✓	✓
Toshiba Home Computer HX-10 Sample Program	100	✓	✓	✓
Trailblazer	100	✓	✓	✓
Trantor, The Last Stormtrooper (Spanish)	100	✓	✓	✓
Trantor, The Last Stormtrooper (English)	100	✓	✓	✓
Trantor, The Last Stormtrooper (English, Alternate)	100	✓	✓	✓
Tres Luces de Glaurung, Las	100	✓	✓	✓
Tuareg	1	✓	✓	✓
Turbo Girl	100	✓	✓	✓
Tutor (Side A)	100	✓	✓	✓
Tutor (Side B)	100	✓	✓	✓
Uchi Mata (Buggy Version)	100	✗	✓	✓
Uchi Mata (Bugs Fixed)	100	✗	✓	✓
Ulysses	100	✗	✓	✓
Vacuumania	100	✓	✓	✓
Valkyr (Clam Case)	100	✓	✓	✓
Valkyr (Jewel Case)	100	✓	✓	✓
Vampire	0	✗	✗	✗
Vestron	100	✓	✓	✓

Viaje al Centro de la Tierra	100	✓	✓	✓
Viaje al Centro de la Tierra Ed.Ext.	100	✓	✓	✓
Void Runner	100	✓	✓	✓
Vortex Raider	100	✓	✓	✓
WEC Le Mans	100	✓	✓	✓
The Way of the Tiger (Side A)	1	✓	✓	✓
The Way of the Tiger (Side B)	1	✓	✓	✓
Wham! - A caixa de Musica	0	✗	✗	✗
Who Dares Wins II	100	✓	✓	✓
Whopper Chase	100	✓	✓	✓
Xenon	100	✗	✓	✓
Xybots	1	✓	✓	✓
Xyxolog	100	✓	✓	✓
Zakil Wood	100	✗	✓	✓
Zanac	100	✓	✓	✓
Zaxxon (requires 32k in first slot)	100	✓	✗	✗
Zipi y Zape Siade A)	100	✓	✓	✓
Zipi y Zape Siade B)	100	✓	✓	✓
Zoids - The Battle Begins	100	✓	✓	✓
Zond - The Final Combat	100	✓	✓	✓

It should be noted that even as a standard, MSX does not provide perfect software compatibility between machines, especially those with 32K base memory or less even when a memory expansion cartridge is installed.

APPENDIX B. IMAGE LOADING ISSUES

There are no guarantees in life and likewise the author of **SVI-CAS** cannot claim that **SVI-CAS** will play-back every image you may find on the web. In particular, during the development of this product more than 1,000 images of various supported formats have been tested and hundreds internally inspected leading to the observation that many people who create such files do not always follow the various format standards accurately – the worst offenders being the creators of TRS-80 CoCo and Dragon **CAS** images, many of which lack either leader or trailer bytes throughout an image in the desire to optimize them for storage and emulator use. In the case of SVI-318/328 **CAS** images it was noted that some images lack leader tone bytes under the guise of also saving image space.

SVI-CAS will attempt to rebuild on-the-fly when playing such an image by performing byte stuffing for packet reconstruction or simply provision the missing leader when playing said blocks, so the host computer will be none the wiser. A small number of TRS-80 CoCo and Dragon **CAS** images are a mixed situation where they commence as a standard image, but then continue on in a custom format – preserving the original tape protection scheme. In such as instance, if **SVI-CAS** detects such a change it will switch to “raw mode” reproduction in an attempt to continue to play the image without interruption,, one such image requiring this tactic for correct reproduction is the CoCo **CAS** file “Zaxxon”. In general **SVI-CAS** will achieve meaningful playback of most Dragon **CAS** images, though there are a small number archived within the Dragon TOSEC collection that are so heavily modified they prove impossible to reconstruct on the fly during playback.

When using third party software tools to create MSX **CAS** images it is important to ensure that each new data block commences at an address in the image file that is an exact multiple of eight bytes, there are at least two utilities available on the web that the author is aware of that fail to enforce this **CAS** requirement. **SVI-CAS** will not play images properly that contain this flaw.

Sinclair computers have a rather primitive tape interface when compared to other machines of the 1980s and can prove unreliable – especially where a custom high speed loading protection system is part of a tape image. If issues are encountered it is suggested to retry loading an image and observe the behaviour of the computer. Experimentation in the form of manually extending the pause time between program parts may help for example – where a program has not had enough time to prepare to load the next part before actual playing of the next part begins. The author has encountered several tape images where this is the case.

Additionally, some ZX Spectrum programs when loading look like they have actually failed when everything is going well, for example the game “Pacland” will present a solid white border with black screen – giving no indications loading is proceeding till the game image appears, while “Lords of the Ring” will skip several large junks of data in the middle of the tape image while loading, again giving the appearance that something is wrong – so don’t be fooled! If an image file is not playing back correctly and / or the host computer is failing to load, the following basic checks should be made before examining the image its self:

- Has the correct interface been selected ?
- Does this image require a specific baud rate (custom loader) to work ? Is it too high ?
- Are the connections between **SVI-CAS** and the host computer sound ? – an intermittent tape motor control connection can wreak havok during play-back.
- Is the loading of this image phase sensitive? Try setting this attribute to opposite, some machines tolerate both phases at low baud rates, bur become fussy at higher baud rates.

- For Commodore machines has the user selected the correct video standard **PAL/NTSC** ?
- Does the image require automatic play-back control (tape motor via computer) as opposed to manual manipulation in order to satisfy a custom loader ?
- For Atari machines has the user set a too higher baud rate for their particular model?
- For some MSX machines, 3600 baud is not natively supported, so drop the selected baud rate to 2400.

When first powering up, **SVI-CAS** factory defaults to known settings that work with most images used by the author at the time of writing this manual. If the user makes any changes, these are automatically saved and will be the defaults presented when the unit is re-powered at a future point in time.

It should be noted that general loading behaviour between the different supported computers does vary, some are more informative of loading progress while others only inform the user at the end of the loading progress, as the user's experience with using **SVI-CAS** grows they will become more proficient at selecting the most appropriate options for optimal image loading for a specific image type, program, and computer.

Commodore TAP images are raw digital recordings of real analog tape playbacks – this means that not only do they preserve the analog tape signal, but any noise and quality issues along with that signal. As a result, some raw TAP images for this family of machines will be encountered whose quality is low as to be problematic in terms of loading reliability, the user is suggested to enable audio playback during some sample loadings of Commodore TAP images in order to hear the varying quality of TAP recordings for Commodore computers, often audio feedback and general background noise is quite audible and may represent the source of loading issues for a particular image.

In a similar respect the quality of wave images will also affect the ease or even just the possibility of successful loading, if difficulty is being experienced when attempting to load a wave image it can be beneficial to check the following:

Does the image conform to **SVI-CAS** requirements in terms of image attributes (see listed specifications in section 12) ?

Does the image contain a poor quality signal that exhibits a high content of noise ?

Does the image contain sections where the recorded signal is damaged or actually missing or is of excessively low amplitude ?

Problems such as those suggested above can be readily observed if the problem wave image is loaded into a tool such as “Audacity”. Such tools also offer capabilities that allow the user to attempt to improve / repair / or re-sample a wave image which may help with loading and allow meeting of the **SVI-CAS** requirements.

APPENDIX C. GENERIC MODE SETTINGS

The following table lists additional vintage computers that have been tried successfully with **SVI-CAS** and lists their playback and recording settings required to correctly save and load programs with this unit when the Generic Mode interface selection is made, the general rule is when the host machine supports tape motor control the user should make use of that functionality.

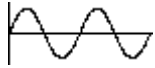
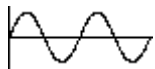
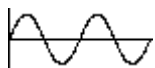
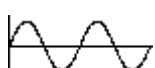
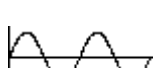
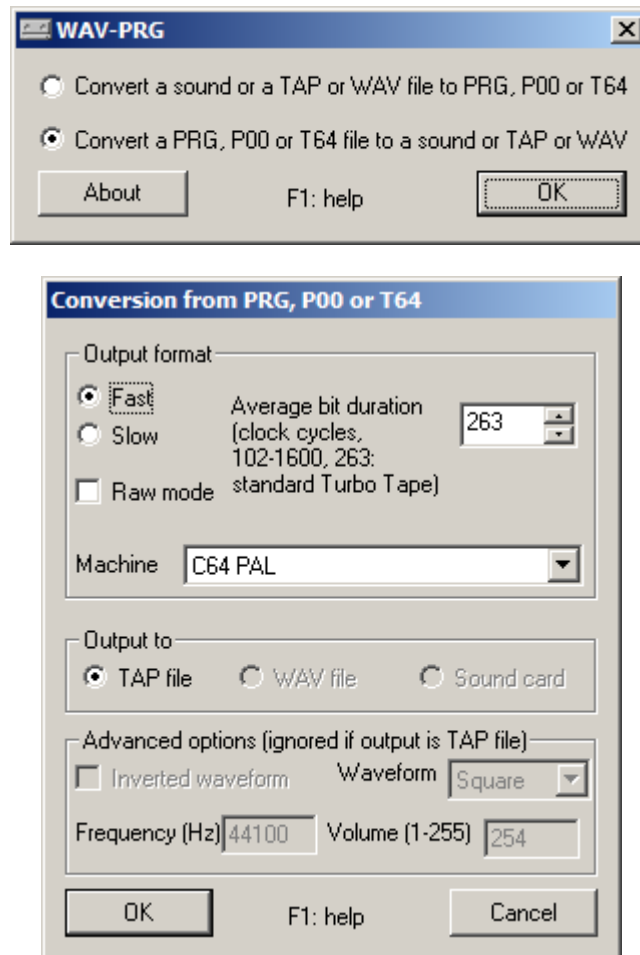
Vintage System	Playback Phase	Recording Trigger	Tape Motor Control
TRS80 Model 1			 Enabled
TRS80 Model 3			 Enabled
TRS80 Model 4			 Enabled
Apple II/II+			Free Running
Apple IIe			Free Running
Microbee			 Enabled
SEGA SC-3000			Free Running

Table of Generic Settings for Known Host Computer(s)

APPENDIX D. GENERATING C16 / C64 / VIC20 TAPs

For Commodore users there exists an excellent utility called **WAV-PRG**, this utility allows the user to convert **PRG**, **P00**, or **T64** images to a native **TAP** image which then may be used with **SVI-CAS**.



Sample Conversion Process Screen(s)

When converting such iamges for the Commodore 64 it even allows a turbo tape loader to be added to the start of a generated **TAP** image allowing far faster loading of the converted program(s). The utility is feature packed and allows the user to experiment with loading speed settings which can take advantage of **SVI-CAS** being a digital device rather than an analog tape recorder.

The latest version of **WAV-PRG** may be downloaded at its Sourceforge home page which is:

<https://sourceforge.net/projects/wav-prg/>

APPENDIX E. WHERE CAN I FIND CASSETTE IMAGES ?

There are many active archives on the web preserving software for the machines supported by **SVI-CAS**, some suggested places to try are:

For the Spectravideo SVI-318 / 328 computers:

The Old Computer archives:

[https://www.theoldcomputer.com/roms/index.php?folder=Spectravideo/SVI-318-SVI-328/Various/\[CAS\]](https://www.theoldcomputer.com/roms/index.php?folder=Spectravideo/SVI-318-SVI-328/Various/[CAS])

Samdal :

<http://www.samdal.com/svsoftware.htm>

TOSEC Spectravideo Archive:

https://archive.org/details/Spectravideo_SVI-318_SVI-328_TOSEC_2012_04_23

TSX Files Repository:

<https://tsx.eslamejor.com/>

For MSX computers:

Planet Emulation:

<http://www.planetemu.net/machine/msx-1-2>

TSX Files Repository:

<https://tsx.eslamejor.com/>

TOSEC MSX Archive:

https://archive.org/details/MSX_MSX_TOSEC_2012_04_23

For the TRS-80 Color Computer:

TRS-80 Color Computer Archive:

<https://colorcomputerarchive.com/>

For the Dragon 32 / 64 Computer:

World of Dragon:

<http://archive.worldofdragon.org/>

For the Sinclair ZX80:

Sinclair ZX80 Resource Centre:

https://fruitcake.plus.com/Sinclair/ZX80/ZX80_ResourceCentre.htm

For the Sinclair ZX81:

Sinclair ZX81 Resource Centre:

http://www.fruitcake.plus.com/Sinclair/ZX81/ZX81_ResourceCentre.htm

Sinclair ZX81 Download Page:

<http://www.zx81.nl/>

ZX81 Stuff:

<https://www.zx81stuff.org.uk/>

TOSEC ZX81 Archive:

https://archive.org/details/Sinclair_ZX81_TOSEC_2012_04_23

For the Sinclair ZX Spectrum:

World of Spectrum:

<https://worldofspectrum.org/>

The TZX vault:

<https://tzxvault.org/>

TOSEC ZX Spectrum Archive:

https://archive.org/details/Sinclair_ZX_Spectrum_TOSEC_2012_04_23

Planet Emulation:

<http://www.planetemu.net/machine/zx-spectrum>

For the Amstrad CPC:

CPC Power:

<https://www.cpc-power.com/index.php?page=jeux>

The TZX vault:

<https://tzxvault.org/>

For the Acorn BBC Micro & Electron:

Stairway To Hell:

<http://www.stairwaytohell.com/>

For the Commodore VIC-20:

Armas VIC-20 TAP files & Cassettes:

<http://armas.cbm8bit.com/vicimages.htm>

TOSEC VIC-20 Archive:

https://archive.org/details/tosec-20161111-commodore-vic-20_201703

For the Commodore 64/128:

Planet Emulation:

<https://www.planetemu.net/machine/commodore-64>

TOSEC Commodore 64 Archive:

https://archive.org/details/Commodore_C64_TOSEC_2012_04_23

For the Oric-1 & Oric Atmos:

Planet Emulation:

<https://www.planetemu.net/machine/oric>

For the Atari Family:

Atarimania:

<http://www.atarimania.com/>

APPENDIX E. OLDER SVI-CAS HW & SW UPGRADING

Owners of older units may at some point desire to bring their unit up to the current specification, I generally offer a free service including the parts that must be added to the unit and my labor, the owner of the unit though will have to pay for postage. If you have an older unit and feel the postage costs are worth it, you may contact me vis the email address below to arrange for this to be done, please note that extra cables for Spectravideo and Commodore machines would need to also be paid for.

Revision 1, 1.5 & 2.0 are upgradable to 3.0, unfortunately the hardware of units that can record is significantly different to those older units that only playback and therefore not a possible upgrade option.

The answer to the question, why isn't this unit self-upgradable, the answer is simply author desires to protect the IP of the unit and that the later revisions require specific modifications to the hardware that if incorrectly carried out may damage the unit or host computer(s).

To arrange an upgrade please email: svi_projects@protonmail.com

APPENDIX F. SVI-CAS DIRECT PRICING

The best way to buy **SVI-CAS** and/or the adapter cables at the lowest possible price is to deal directly with the author of this equipment. Although units appear on **ebay** from time to time or other online outlets, the user will always pay less buying direct and can specify the preferred colour scheme of their unit rather than simply accepting what is on offer.

The author accepts **PayPal** for payments – so the purchaser has the same level of buyer protection as if they purchasing from EBay or any other online retail site. All purchases are shipped only via tracked airmail.

All sales come with a 90 day warranty on parts and labor – buyer pays postage, the author will either repair or replace the unit at his discretion.

General Price List as of May 2022:

SVI-CAS	\$ 95.00 Australian
SVI Adapter Cable	\$ 15.00 Australian
Commodore Adapter Cable	\$ 15.00 Australian
Atari Adapter Cable	\$ 20.00 Australian

PayPal is accepted.

To order please email: svi_projects@protonmail.com